

Malawi University of Science and Technology

Malawi Institute of Technology

PHYS-111:

Mechanics and Thermal Properties of Matter

Part 1: Notes



Unit 1

Units and Measurement

1.3 Measurement and unit

1.3.1 Physical quantity

A physical quantity is any quantity that can be measured e.g length, time, speed, density, electric charge, temperature

Units: A physical quantity is expressed as a number with units e.g length: $L=2\text{m}$, time; $t=1\text{s}$. Meter represented by m is a unit of length and second represented by s is unit of time.

A unit is a standard quantity (unit comes from latin for 'one') e.g length: 1 meter, time: 1 second, mass 1 kilogram.

SI Units

SI stands for *systeme internationale* in French, meaning “international system”. Therefore, SI units are agreed international standard units. Base quantities are measured using SI units.

Basic physical quantities are quantities which are not expressed in terms of other quantities. They are also called fundamental physical quantities.

The SI units of base quantities are called basic units or fundamental units.

There are seven base quantities as shown in Table 1 below:

Table 1: Base Physical Quantities

Quantity	SI Unit Name	Unit Symbol
Mass	kilogram	kg
Length	meter	m
Time	second	s
Electric current	Ampere	A
Temperature	Kelvin	K

Amount of substance	mole	mole
Luminous intensity	candela	cd

Derived quantities are expressed in terms of basic quantities by multiplication or division. Table 2 shows some examples of derived quantities.

Table 2: Examples of Derived Physical Quantities

Derived Quantity	Unit Name	Unit Symbol
Force	Newton	$N = \text{kgms}^{-2}$
Acceleration	Meter per second square	ms^{-2}
Pressure	Pascal	$\text{Pa} = \text{Nm}^{-2}$
Work	Joule	$J = \text{kgm}^2$
Power	Watt	$W = \text{kgm}^2\text{s}^{-3}$
Density	Mass per volume	kgm^{-3}

Unit Prefixes

For larger and smaller units, we write them in a short hand by multiply by powers of 10 as in Figure 3.

Figure 3: Unit Prefixes

Prefix	Symbol	Factor	Prefix	Symbol	Factor
Peta	P	10^{15}	deci	d	10^{-1}
Tera	T	10^{12}	centi	c	10^{-2}
Giga	G	10^9	milli	m	10^{-3}
Mega	M	10^6	micro	μ	10^{-6}
Kilo	k	10^3	nano	n	10^{-9}
Hector	h	10^2	pico	p	10^{-12}
Deca	da	10	fento	f	10^{-15}

Examples: $\text{cm} = 0.01\text{m} = 10^{-2} \text{ m}$, $\text{mm} = 0.001 \text{ m} = 10^{-3} \text{ m}$, $\text{km} = 1000 \text{ m} = 10^3 \text{ m}$

Changing units

Replace the unit you are given with the equivalent quantity in the unit you want

Example : Write 72km/hr in m/s

Solution

We know that $1\text{km} = 1000\text{m}$ (we are given km, we want m)

We also know that $1\text{hr} = 3600\text{s}$ (we are given hr, we want s)

$$\text{Then: } 72 \frac{\text{km}}{\text{hr}} = 72 \times \frac{1000 \text{ m}}{3600 \text{ s}} = 20 \text{ m/s}$$

SI standards

Length

Length is defined as the distance between two points. The SI unit of length is meter.

One standard meter is equal to 1 650 763.73 wavelengths of the orange – red light emitted by the individual atoms of krypton – 86 in a krypton discharge lamp.

Mass

Mass is the quantity of matter contained in an object? It is independent of temperature and pressure. It does not vary from place to place. The SI unit of mass is kilogram.

One kilogram is equal to the mass of the international prototype of the kilogram (a platinum-iridium alloy cylinder) kept at the International Bureau of Weights and Measures at Sevres, near Paris, France.

Time

Until 1960 the standard of time was based on the mean solar day, the time interval between successive passages of the sun at its highest point across the meridian. It is averaged over an year. In 1967, an atomic standard was adopted for second, the SI unit of time.

One standard second is defined as the time taken for 9 192 631 770 periods of the radiation corresponding to unperturbed transition between hyperfine levels of the

ground state of cesium – 133 atom. Atomic clocks are based on this. In atomic clocks, an error of one second occurs only in 5000 years.

Activity 1.1

1. Define physical quantity
2. Explain the difference between basic quantities and derived quantities
3. Classify each of the following quantities as a basic quantity or a derived quantity:
 - a. $30\mu s$
 - b. $20ms^{-2}$
 - c. 1000 kJ
 - d. 5W

1.3.2 Dimensional analysis

It deals with the physical quantity in question. It describes any quantity in terms of fundamental quantities.

Dimensions of a physical quantity are the powers to which the fundamental quantities must be raised. Table 4 shows some quantities with their dimensions.

Table 4: Some Physical quantities with their dimensions

Quantity	Dimension
Mass	M
Length	L
Time	T
Acceleration	LT^{-2}

We write [X] meaning “dimension of quantity X”

Activity 1.2

1. What are the dimensions of the following physical quantities:
 - a. Area of a triangle
 - b. Stress
 - c. Frequency
 - d. Impulse

Application of Dimensional analysis

Checking if an equation is consistent/valid

An equation is said to be dimensionally consistent if the SI unit(s) on the left side are the same as the SI units on the right of the equation

Example

Let us consider the following equation of motion with constant acceleration:

$$s = ut + \frac{1}{2}at^2$$

where:

s: displacement with dimension of L and SI unit of m

u: velocity with dimensions of LT^{-1} and SI unit of ms^{-1}

a: acceleration with dimensions of LT^{-2} and SI units of ms^{-2}

t: time with dimension of T and SI unit of s

Then, we rewrite the equation, all quantities replaced by their dimensions, we get:

$$s = ut + \frac{1}{2}at^2 \rightarrow L = LT^{-1} \times T + LT^{-2} \times T^2$$

$$L = L + L$$

$$L = L$$

We finally get the same dimension of length both sides of the equation. Therefore, the equation is dimensionally correct.

We also use dimensional analysis to derive an equation

If we know that a quantity is related in same way to other quantities, then it is possible to derive an equation expressing the relationship.

Example: Suppose we are told that the acceleration **a** of a particle moving with uniform speed **v** in a circle of radius **r** is proportional to some power of **r**, say r^n , and some power of **v**, say v^m . Determine the values of **n** and **m** and write the simplest form of an equation for the acceleration.

We write the relation as

$$a \propto r^n v^m$$

$$a = k r^n v^m$$

where **k** is the proportionality constant

Writing the equation dimensionally, we get

$$LT^{-2} = k[L]^n[LT^{-1}]^m$$

k is dimensionless, so we can take it out of the equation

$$LT^{-2} = [L]^n[LT^{-1}]^m$$

$$LT^{-2} = L^n L^m T^{-m}$$

$$LT^{-2} = L^{n+m} T^{-m}$$

equating the powers, we get

$$1 = n + m \text{ and } -2 = -m$$

from which, we get $m = 2$ and $n = -1$

Therefore, the derived equation is

$$a = k \frac{v^2}{r}$$

Activity 1.3

1. Given that, the time period *T* of a simple pendulum depends upon mass *m* of the bob, length *l* of the pendulum and acceleration due to gravity *g* at the place where the pendulum is suspended. Find an expression for the time period **T** in terms of **m**, **l** and **g**.

1.3.3 Decimal point, scientific notation, order of magnitude, precision, accuracy and order of magnitude

Decimal point: the number 7.32 has two decimal places and 1500 has no decimal point.

Scientific notation: this is writing numbers in powers of 10. In general any number in the form of $x \times 10^y$ is in scientific notation where x is any number such that $1 \leq x \leq 10$ and y is an exponent. E.g 7.41×10^3 is in scientific notation.

Example

Convert 6342147.14 to scientific notation

Solution

$$6.3421714 \times 10^6$$

When counting decimal places from right to left, the exponent is positive

Example

Write 0.0042 in scientific notation

Solution

$$4.2 \times 10^{-3}$$

When counting decimal places from left to right, the exponent is negative

Order of magnitude: it is a rough approximation of a quantity to the closest power of 10 e.g $8 \sim 10$ this means 8 is of the order of 10

Examples

$123 \sim 100$, and $0.07 \sim 0.1$

To find order of magnitude, round up or down to nearest power of 10 e.g $7.62 \times 10^4 \sim 10^5$, $3.2 \times 10^2 \sim 10^2$, $8.417 \times 10^{-6} \sim 10^{-5}$, $1.5 \times 10^{-8} \sim 10^{-8}$

Significant figures (s.f) are the number of digits in scientific notation e.g 4.71 has 3 s.f.

Uncertainty in measurement: in any measurement there is always an error, and this is called uncertainty.

Accuracy refers to closeness of a measured value to a standard or known value. For example, if in a lab you obtain a weight measurement of 3.2 kg for a given substance, but the actual or the known weight is 10 kg, then your measurement is not accurate. In this case your measurement is not close to the known value.

Precision refers to the closeness of two or more measurements to each other. For example, if you weigh a given substance five times and get 3.2 kg each time, then your measurement is very precise.

Unit 2

Scalar and vector quantities

2.3 Scalar and Vector Quantities

Scalar quantity is a physical quantity that is completely described by magnitude only, e.g mass, temperature, distance and speed. A scalar quantity tells you how much a given quantity is. For example, 20 Kg of meat is a scalar quantity. It is simply telling you how much meat is there. No direction is required for it to make a complete sense.

Vector quantity is a physical quantity that requires the specification of magnitude and direction e.g velocity, displacement and acceleration. A vector tells you how much is the quantity and which way. For example if you drive a bicycle East along the road at 5ms^{-1} . What you have here is a vector-velocity. The bicycle is moving at 5ms^{-1} (this is the magnitude) and you know where you are going –East (this is the direction).

Activity 2.1

1. Explain the difference between vector and scalar quantities
2. Give examples of physical quantities that are both base and scalar
3. Give examples of physical quantities that are both derived and vector

2.3.1 Vector Notation

Mathematical representation: vectors are represented by a symbol with an arrow pointing to the right above it e.g $\vec{a}$, $\vec{v}$ and $\vec{F}$ represents vectors acceleration, velocity and force respectively.

Two vectors $\vec{OA}$ and $\vec{BC}$ are equal if they have the same magnitude and the same direction. This property allows us to translate a vector parallel to itself in a diagram without affecting the vector. See Figure 1.3.1, vector $\vec{OA}$ is translated to vector $\vec{BC}$.

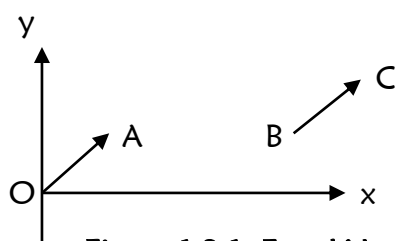


Figure 1.3.1: Equal Vectors

Graphical Representation: Vectors are drawn as arrows. An arrow has both a magnitude (how long it is) and a direction (the direction in which it points). The starting point of a vector is known as the tail and the end point is known as the head as shown in the Figure below:

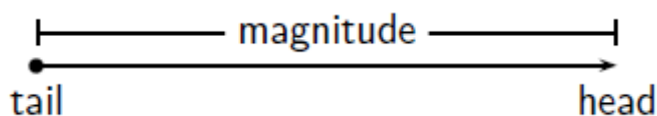
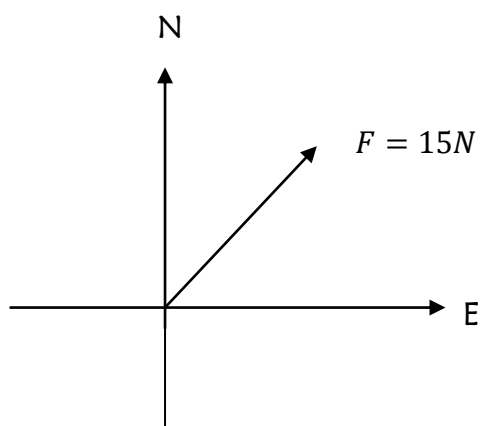


Figure 1.3.2: Parts of a vector

If you want to present a force of 15 N North East, you present it as follows:



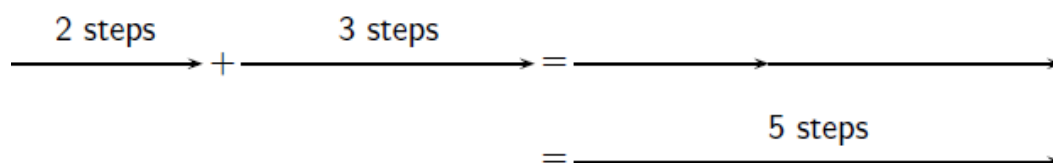
Activity 2.2

1. Represent the following vectors geometrically on a plane:
 - i. Velocity, $v = 10\text{ms}^{-1}$ North
 - ii. Velocity, $v = 10\text{ms}^{-1}$ North West
 - iii. Acceleration $a = 5\text{ms}^{-2}$ at an angle of 300°

2.3.2 Adding vectors

When vectors are added, we need to add both a magnitude and a direction. For example, take 2 steps in the forward direction, stop and then take another 3 steps in the forward direction. The first 2 steps is a displacement vector and the second 3 steps

is also a displacement vector. If we did not stop after the first 2 steps, we would have taken 5 steps in the forward direction in total. Therefore, if we add the displacement vectors for 2 steps and 3 steps, we should get a total of 5 steps in the forward direction. Graphically, this can be seen by first following the first vector two steps forward and then following the second one three steps forward:



The final answer when adding vectors is called the resultant. The resultant displacement in this case will be 5 steps forward.

Definition: Resultant of Vectors

The resultant of a number of vectors is the single vector whose effect is the same as the Individual vectors acting together

2.3.3 Techniques of Vector Addition

1. Graphical Techniques

Graphical techniques involve drawing accurate scale diagrams to denote individual vectors and their resultants.

We next discuss the two primary graphical techniques, the head-to-tail technique and the parallelogram method.

The Head-to-Tail Method

In describing the mathematical properties of vectors we used displacements and the head-to-tail graphical method of vector addition as an illustration. The head-to-tail method of graphically adding vectors is a standard method that must be understood.

Steps to follow: Head-to-Tail Method of Vector Addition

1. Choose a scale and include a reference direction.
 2. Choose any of the vectors and draw it as an arrow in the correct direction and of the correct length – remember to put an arrowhead on the end to denote its direction.
 3. Take the next vector and draw it as an arrow starting from the arrowhead of the first vector in the correct direction and of the correct length.
-

4. Continue until you have drawn each vector – each time starting from the head of the previous vector. In this way, the vectors to be added are drawn one after the other head-to-tail.

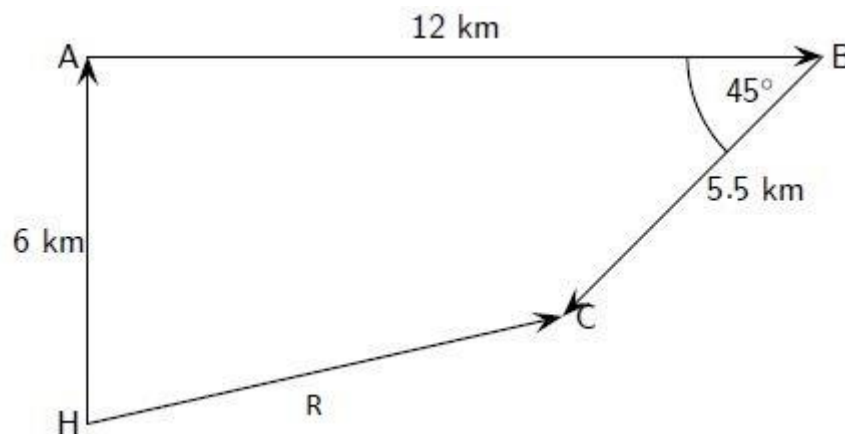
5. The resultant is then the vector drawn from the tail of the first vector to the head of the last. Its magnitude can be determined from the length of its arrow using the scale. Its direction too can be determined from the scale diagram.

Example 2.1

A ship leaves harbor H and sails 6 km north to port A. From here the ship travels 12 km east to port B, before sailing 5,5 km south-west to port C. Determine the ship's resultant displacement using the head-to-tail technique of vector addition.

Solution

Step1: Draw a rough sketch of the situation



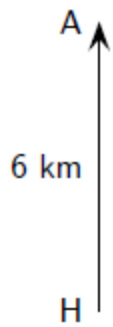
Step 2: Choose a scale and include a reference direction.

The choice of scale depends on the actual question – you should choose a scale such that your vector diagram fits the page.

It is clear from the rough sketch that choosing a scale where 1 cm represents 2 km (scale: 1 cm = 2 km) would be a good choice in this problem. The diagram will then take up a good fraction of an A4 page. We now start the accurate construction.

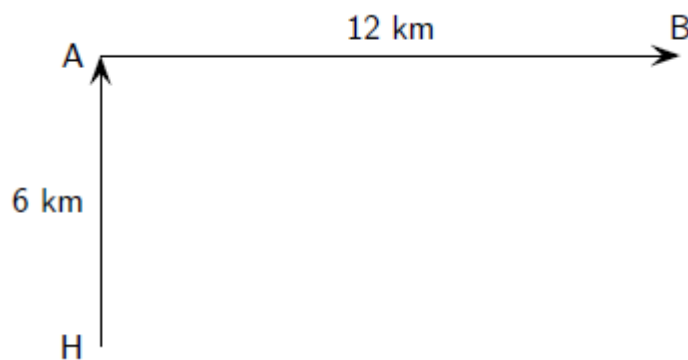
Step 3 : Choose any of the vectors to be summed and draw it as an arrow in the correct direction and of the correct length – remember to put an arrowhead on the end to denote its direction.

Starting at the harbour H we draw the first vector 3 cm long in the direction north.



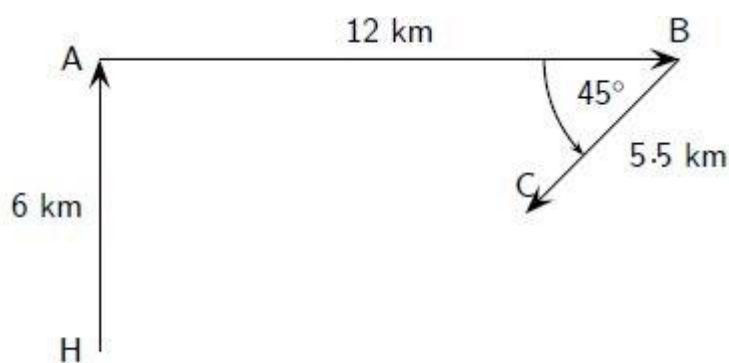
Step 4: Take the next vector and draw it as an arrow starting from the head of the first vector in the correct direction and of the correct length.

Since the ship is now at port A we draw the second vector 6 cm long starting from point A in the direction east.



Step 5: Take the next vector and draw it as an arrow starting from the head of the second vector in the correct direction and of the correct length.

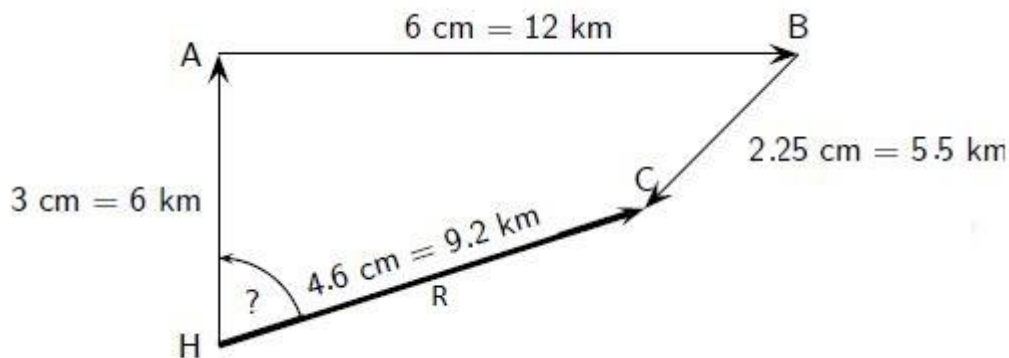
Since the ship is now at port B we draw the third vector 2,25 cm long starting from this point in the direction south-west. A protractor is required to measure the angle of 45° .



Step 6: The resultant is then the vector drawn from the tail of the first vector to the head of the last. Its magnitude can be determined from the length of its arrow using the scale. Its direction too can be determined from the scale diagram.

As a final step we draw the resultant displacement from the starting point (the

Harbor H) to the end point (port C). We use a ruler to measure the length of this arrow and a protractor to determine its direction.



Step 7: Apply the scale conversion.

We now use the scale to convert the length of the resultant in the scale diagram to the actual displacement in the problem. Since we have chosen a scale of $1 \text{ cm} = 2 \text{ km}$ in this problem the resultant has a magnitude of 9.2 km . The direction can be specified in terms of the angle measured either as 072.3° east of north or on a bearing of 072.3° .

Step 8: Quote the final answer

The resultant displacement of the ship is 9.2 km on a bearing of 072.3° .

The Parallelogram Method

The parallelogram method is another graphical technique of finding the resultant of two vectors.

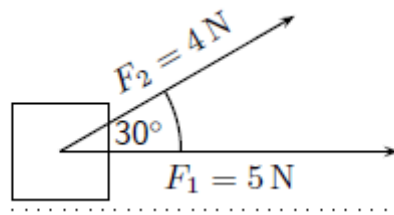
2. The Parallelogram Method

1. Choose a scale and a reference direction.
2. Choose either of the vectors to be added and draw it as an arrow of the correct length in the correct direction.
3. Draw the second vector as an arrow of the correct length in the correct direction from the tail of the first vector.
4. Complete the parallelogram formed by these two vectors.
5. The resultant is then the diagonal of the parallelogram. The magnitude can be determined from the length of its arrow using the scale. The direction too can be determined from the scale diagram.

Example 2.2 Parallelogram Method of Vector Addition

A force of $F_1 = 5\text{ N}$ is applied to a block in a horizontal direction.

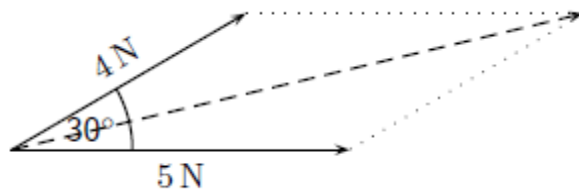
A second force $F_2 = 4\text{ N}$ is applied to the object at an angle of 30° above the horizontal.



Determine the resultant force acting on the block using the parallelogram method of accurate construction.

Solution

Step 1: Firstly make a rough sketch of the vector diagram

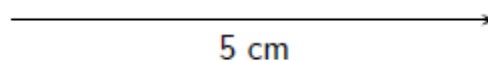


Step 2: Choose a suitable scale

In this problem a scale of $1\text{ cm} = 1\text{ N}$ would be appropriate, since then the vector diagram would take up a reasonable fraction of the page. We can now begin the accurate scale diagram.

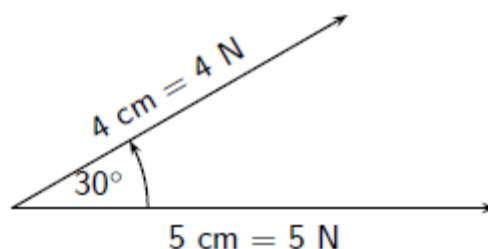
Step 3: Draw the first scaled vector

Let us draw F_1 first. According to the scale it has length 5 cm .



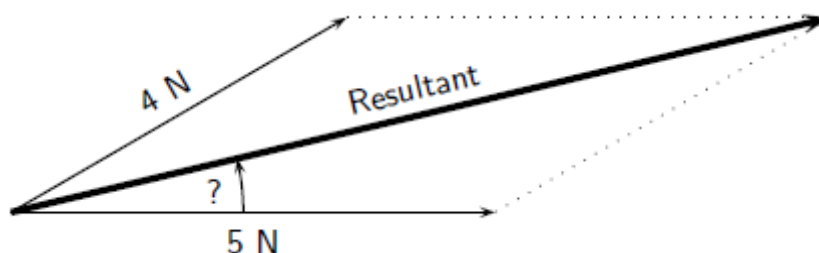
Step 4: Draw the second scaled vector

Next we draw F_2 . According to the scale it has length 4 cm . We make use of a protractor to draw this vector at 30° to the horizontal.



Step 5: Determine the resultant vector

Next we complete the parallelogram and draw the diagonal.



The resultant has a measured length of 8.7 cm.

Step 6: Find the direction

We use a protractor to measure the angle between the horizontal and the resultant.

We get 13.3° .

Step 7: Apply the scale conversion

Finally we use the scale to convert the measured length into the actual magnitude.

Since $1 \text{ cm} = 1 \text{ N}$, 8.7 cm represents 8.7 N. Therefore the resultant force is 8.7 N at 13.3° above the horizontal.

The parallelogram method is restricted to the addition of just two vectors. However, it is arguably the most intuitive way of adding two forces acting at a point.

3. Algebraic Addition and Subtraction of Vectors

Vectors in a Straight Line

Whenever you are faced with adding vectors acting in a straight line (i.e. some directed left and some right, or some acting up and others down) you can use a very simple algebraic technique:

Steps in Addition/Subtraction of Vectors in a Straight Line

1. Choose a positive direction. As an example, for situations involving displacements in the directions west and east, you might choose west as your positive direction. In that case, displacements east are negative.
2. Next simply add (or subtract) the vectors using the appropriate signs.

3. As a final step the direction of the resultant should be included in words (positive results are in the positive direction, while negative resultants are in the negative direction).

When vectors act at an angle, Simple geometric and trigonometric techniques can be used to find resultant vectors.

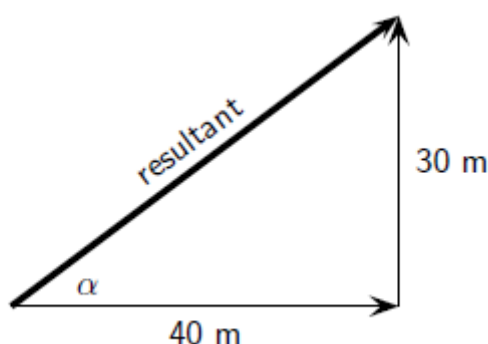
Example 2.3

A man walks 40 m East, then 30 m North. Calculate the man's resultant displacement.

Solution

Step 1: Draw a rough sketch

As before, the rough sketch looks as follows:



Step 2: Determine the length of the resultant

Note that the triangle formed by his separate displacement vectors and his resultant displacement vector is a right-angle triangle. We can thus use the Theorem of Pythagoras to determine the length of the resultant. Let R represent the length of the resultant vector. Then:

$$R^2 = (40\text{m})^2 + (30\text{m})^2$$

$$R = \sqrt{(40\text{m})^2 + (30\text{m})^2}$$

$$\therefore R = 50\text{m}$$

Step 3: Determine the direction of the resultant

Now we have the length of the resultant displacement vector but not yet its direction. To determine its direction we calculate the angle α between the resultant displacement vector and East, by using simple trigonometry:

$$\tan \alpha = \frac{30\text{m}}{40\text{m}}$$

$$\alpha = \tan^{-1}(30\text{m}/40\text{m})$$

$$\alpha = 36.9^\circ$$

Step 4: Quote the resultant

The resultant displacement is then 50 m at 36.9° North of East.

This is exactly the same answer we arrived at after drawing a scale diagram!

2.3.4 Vector Resolution

A vector can be resolved into x-component and y-component. The sum of these components is equal to the original resultant vector.

Vector resolution is the process of breaking a vector into its components. The components of a vector are also vectors and are perpendicular to each other. Vector resolution can be done anywhere in the Cartesian plane as long as the components of the vector are perpendicular to each other.

Consider vector **OA** shown in Figure 2.3 below which is acting at an angle of $\angle AOX$:

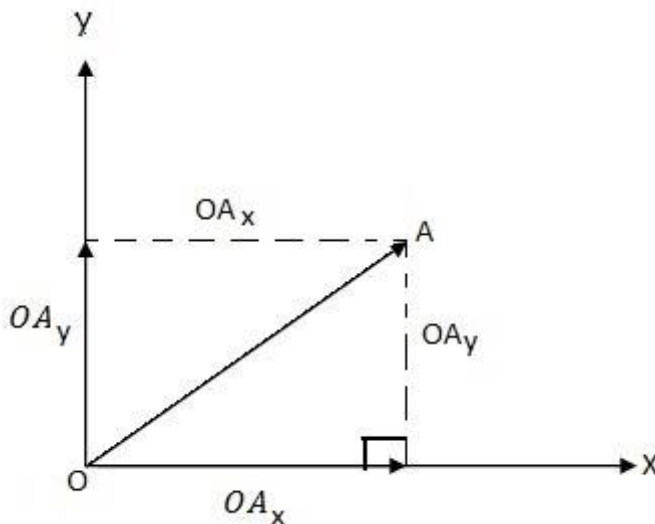


Figure 2.3 Components of Vector OA

In Figure 2.3 OA_x and OA_y are x-component and y-component of vector **OA** respectively.

Using trigonometry, you can get the components of the vector shown in Figure 2.3 by:

$$OA_x = OA \cos(\angle AOX)$$

and

$$OA_y = OA \sin(\angle AOX)$$

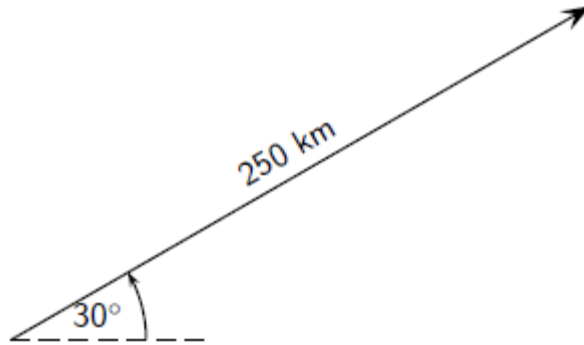
In general for any given vector **Q** at any angle of θ from X-axis, the x-component and y-components can be found as,

$$Q_x = Q\cos\theta \text{ and } Q_y = Q\sin\theta \text{ respectively}$$

Example 2.4

A motorist undergoes a displacement of 250 km in a direction 30° North of East. Resolve this displacement into components in the directions north and east.

Solution

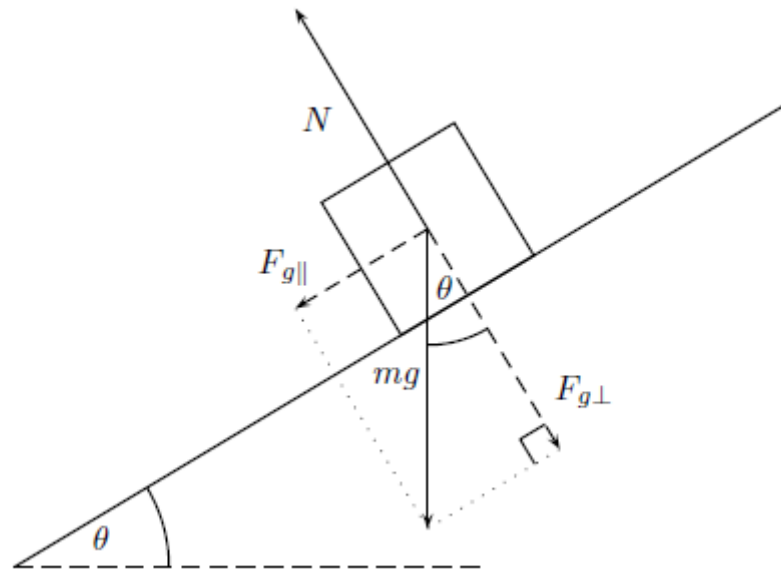


Then Eastern component is the same as x-component $= 250 \cos 30^\circ = 125 \text{ km}$ and North component is the same as y-component $= 250 \sin 30^\circ = 125 \text{ km}$

Example 2.5

As a further example of components let us consider a block of mass m placed on a frictionless surface inclined at some angle θ to the horizontal. The block will obviously slide down the incline, but what causes this motion?

The forces acting on the block are its weight, mg , and the normal force N exerted by the surface on the object and they are equal. These two forces are shown in the diagram below.



Resolving weight into its components, we get:

$$F_{g\parallel} = mg \sin \theta$$

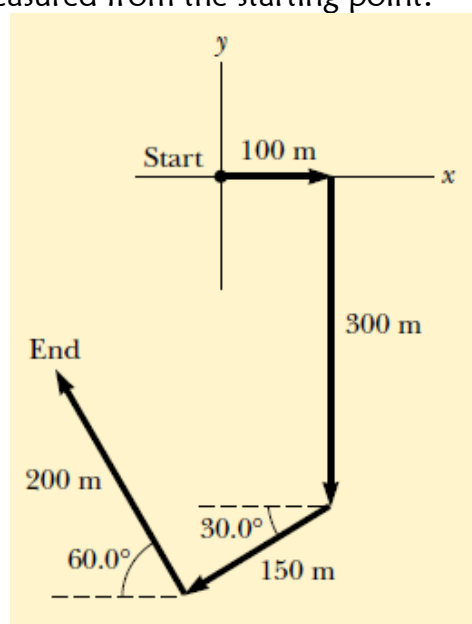
and

$$F_{g\perp} = mg \cos \theta$$

The component of the weight perpendicular to the slope $F_{g\perp}$ exactly balances the normal force N exerted by the surface. The parallel component, however, $F_{g\parallel}$ is unbalanced and causes the block to slide down the slope.

Example 2.6

A person going for a walk follows the path shown in Figure below. The total trip consists of four straight-line paths. At the end of the walk, what is the person's resultant displacement measured from the starting point?

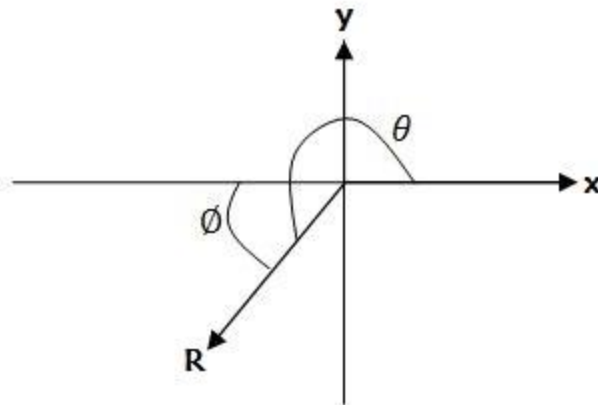


Solution

$$R_x = 100 - 150\cos 30^\circ - 200\cos 60^\circ = -129.90 \text{ m}$$

$$R_y = -300 - 150\sin 30^\circ + 200\sin 60^\circ = -201.79 \text{ m}$$

Then, the resultant displacement can be represented in the Figure below:



$$R = \sqrt{(-129.90)^2 + (-201.79)^2} = 239.99 \text{ m}$$

$$\phi = \tan^{-1}\left(\frac{-201.79}{-129.90}\right) = 57.23^\circ$$

$$\theta = 180^\circ + 57.23^\circ = 237.23^\circ$$

The resultant displacement is 239.99 m at an angle of 57.24° South West or at 237.23°.

Unit 3

Motion in one Dimension

3.3 Kinematics in one dimension

Kinematics is the study of motion of an object. In this unit, you are going to learn motion of an object without considering its cause. The motion of an object can be described by displacement, velocity, acceleration and time. Recall that displacement, velocity and acceleration are vectors. They have both magnitudes and directions. This means that vector methods are employed when dealing with displacement, velocity and acceleration. To do so, you need a frame of reference.

3.3.1 Frame of Reference

A frame of reference is a reference point combined with a set of direction. For example, a boy is standing still inside a train as it pulls out of a station. You are standing on the platform watching the train move from left to right. To you it looks as if the boy is moving from left to right, because relative to where you are standing (the platform), he is moving. According to the boy, and his frame of reference (the train), he is not moving.

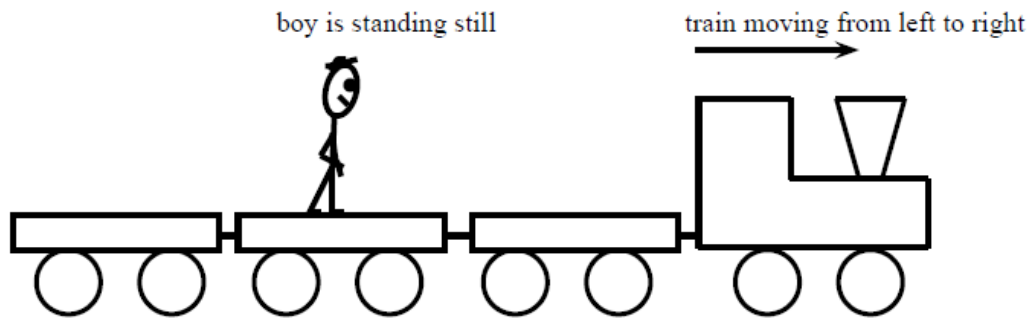


Figure 2.3.1 Frame of Reference

From your frame of reference the boy is moving from left to right.

Displacement is distance covered in a specified direction. It is measured in meters. E.g. 50 m West. Displacement is commonly denoted by letter, s . Figure 3.3.1 Illustrates displacement.

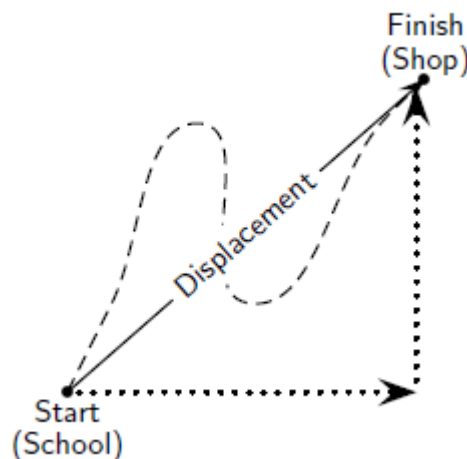


Figure 3.3.1 Illustrates displacement

From Figure 3.3.1, if you walk from school to the shop following the dotted path, then the displacement is the straight line joining the two points from school to the shop. The length of the dotted line gives you the distance.

Distance is the total length between two points. The magnitude of displacement gives you a distance. Distance is also measured in meters, m.

Velocity is the time rate of change of displacement (change in displacement divide by time). It is measured in meters per second (m/s). It is commonly denoted by u for initial velocity and v for final velocity.

Instantaneous velocity is the time rate of change of displacement at an instant time, t . (Change of displacement per infinitesimal time interval, limit of displacement as time approaches to zero i.e

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t}$$

Average velocity is change of displacement per unit time (total displacement divide by total time taken)

Acceleration a , is the time rate of change of velocity. It is measured in (ms^{-2}).

Average acceleration is change of velocity per unit time (total velocity divide by total time taken)

Instantaneous acceleration is the time rate of change of displacement at an instant time, t . (Change of velocity per infinitesimal time or limit of change of velocity as time approaches to zero i.e

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t}$$

3.3.2 Motion graphs

Motion can be presented by a displacement-time graph and velocity-time graph.

3.3.2.1 Displacement-Time graph

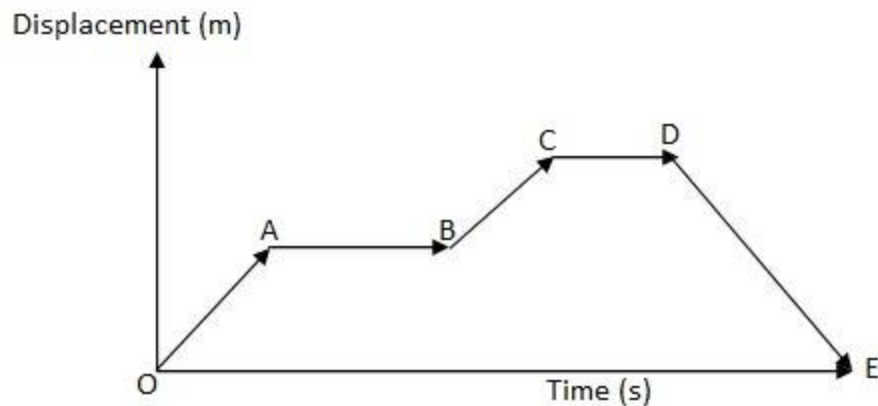


Figure 3.3.3. Displacement-Time graph

In a displacement time graph, the gradient (slope) of the graph gives you the velocity.

From figure 3.3.3, The object is starting from rest and moving with constant velocity between O and A and its velocity is the slope of line segment OA. Between A and B, the object is not moving and its velocity is zero. The motion of an object between B and C and C and D are the same as OA and AC respectively. Between D and E the slope is negative meaning that its velocity is negative. The object is moving back to the original position.

3.3.2.2 Velocity-Time graph

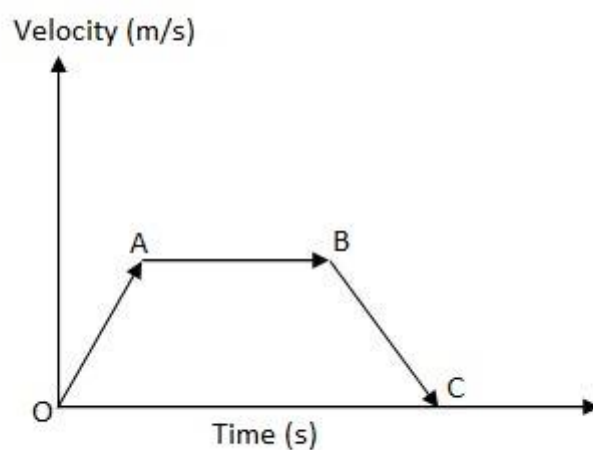


Figure 3.3.4

The slope of Velocity-Time graph gives you acceleration.

In Figure 3.3.4, the object moves with uniform acceleration between OA. This means that its velocity increases uniformly. Between A and B, the object moves with zero acceleration since its velocity is not changing (it is constant). Between B and C, the slope is negative. The object is moving with constant negative acceleration (Deceleration). The velocity between BC is decreasing uniformly.

3.3.3 Equations of Uniform motion (Motion with constant acceleration)

Consider an object which is initially moving at velocity u when you first see it (at $t = 0$). Then during time interval, t , the object increases its velocity uniformly to final value, v . During the same interval its displacement increases from zero to s .

The object's motion is represented by the graph in Figure 3.3.5 below:

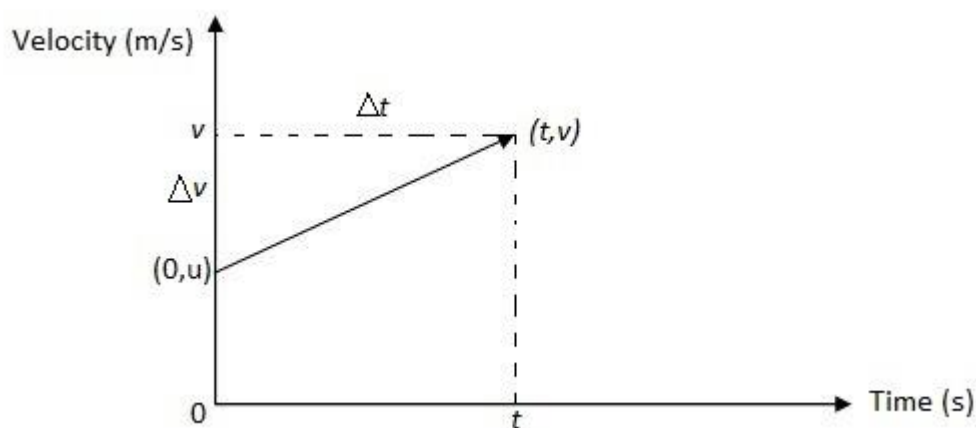


Figure 3.3.5

Recall that acceleration is equal to the gradient of velocity-time graph.

from Figure 3.3.5

$$\text{acceleration} = \frac{\text{Final velocity} - \text{Initial velocity}}{\text{time interval}}$$

$$a = \frac{\Delta v}{\Delta t}$$

$$a = \frac{v - u}{t}$$

$$\therefore v = u + at \quad (3.01)$$

This is the first equation of motion.

The area under velocity-time graph gives you displacement covered by an object during the time interval, t . From the graph in Figure 3.3.5, displacement (s) is equal to the area under the trapezium. Therefore,

$$s = 1/2(u + v)t$$

$$\therefore s = \frac{u+v}{2}t \quad (3.02)$$

This is the second equation of motion.

Substituting equation 3.3.1, into equation 2.3.2 for v , we have:

$$s = \frac{(u + at + u)t}{2}$$

This reduces to:

$$s = ut + \frac{1}{2}at^2 \quad (3.03)$$

This is the third equation of motion.

Making t the subject of the formula in equation 3.3.1 you get:

$$t = \frac{v-u}{a} \quad (3.04)$$

Substituting equation 3.04 into equation 3.02 for t , you get the last equation of motion:

$$s = \frac{v + u}{2} \times \frac{v - u}{a}$$

$$s = \frac{v^2 - u^2}{2a}$$

$$\therefore v^2 = u^2 + 2as \quad (3.05)$$

You will be using these equations to solve problems of motion with constant velocity. For free-fall motion i.e. motion under the influence of gravity, you substitute acceleration, a , with acceleration due to gravity, g .

Example 3.1

A race car starting from rest accelerates at a rate of 5.00 m/s^2 . What is the velocity of the car after it has travelled 100 ft? Given that $1 \text{ m}=3.281 \text{ ft}$.

Solution

$$1\text{m}=3.281 \text{ ft} \rightarrow 1\text{ft}=0.305 \text{ m}$$

Thus, 100 ft = 30.50 m

We have been given

$U=0$ m/s, $a=+5$ m/s², $x=30.5$ m, $v=?$

We apply the following equation:

$$v^2 = u^2 + 2a\Delta x$$

$$v^2 = 0^2 + 2(5\text{m/s}^2)(30.5\text{ m})$$

$$v = \mp\sqrt{305\text{ m}^2/\text{s}^2} = \mp 17.5\text{ m/s}$$

The car is travelling to the right, we choose +17.5 m/s as the correct solution for final velocity v .

Example 3.2

A car travelling at a constant speed of 30.0 m/s passes a trooper hidden behind a billboard. One second after the speeding car passes the billboard, the trooper sets off in chase with a constant acceleration of 3.00 m/s². How long does it take the trooper to overtake the speeding car?

Solution

The car is travelling at a constant speed, therefore, its acceleration is zero and

$$\Delta x = v_i t + \frac{1}{2}at^2$$

$$x_c = x_i + v_i t + \frac{1}{2}at^2$$

$$x_c = 30 + 30t$$

For the trooper who starts from rest i.e $t_i = 0$, $x_i = 0$, $v_i = 0$ and $a = 3\text{m/s}^2$, we have

$$x_T = x_i + v_i t + \frac{1}{2}at^2$$

$$x_T = \frac{1}{2}(3.00\text{ m/s}^2)t^2$$

The trooper overtakes the car at the instant that $x_c = x_T$

$$30 + 30t = \frac{1}{2}(3.00)t^2$$

This reduces to:

$$t^2 - 20t - 20 = 0$$

$$t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$t = \frac{-(-20) \pm \sqrt{(-20)^2 - 4 \times 1 \times (-20)}}{2 \times 1}$$

From which we get:

$$t = 21.0s \text{ or } t = -0.95s$$

Time is never negative. Therefore, the trooper overtakes the car at $t = 21.0 \text{ s}$.

2.3.4 Freely falling object

A freely falling object is the one moving under the influence of gravity only regardless of its initial motion.

Objects thrown upward or downward and those released from rest are all falling freely once they are released.

Once they are in free fall, all objects have acceleration downward, which is free-fall acceleration g .

The equations describing free fall motion are the same as those of an object moving with constant acceleration along x -axis with the substitutions $x = y$ and $a = g$.

For a downward motion $a = g$, because the object moves in the direction of the gravity.

For upward motion $a = -g$, since the object moves against the gravity.

For a freely falling object, $a = g$ and $v_i = 0$, since it starts from rest.

Example 3.3

A golf ball is released from rest at top of a very tall building. Neglecting air resistance, calculate the position and velocity of the ball after 1.00 s .

Solution

Taking the downward motion to be positive we have:

$$y_i = 0 \text{ at } t = 0, v_i = 0, a = g = 9.8 \text{ m/s}^2$$

$$\Delta y = v_i t + \frac{1}{2} a t^2$$

$$y = \frac{1}{2}(9.8)(1.00)^2$$

$$y = +4.90 \text{ m}$$

$$v_f = v_i + at$$

$$v_f = 0 + 9.8 \text{ m/s}^2(1.00\text{s})$$

$$\therefore v_f = 9.8 \text{ m/s}$$

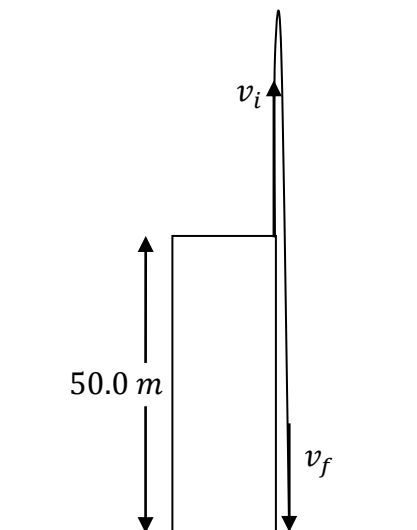
Example 3.4

A stone is thrown from the top of a building with an initial velocity of 20 m/s straight upward. The building is 50.0 m high, and the stone just misses the edge of the roof on its way down. Determine:

- (a) The time needed for the stone to reach its maximum height
- (b) The maximum height
- (c) The time needed for the stone to return to the level of the thrower
- (d) The velocity of the stone at this instant
- (e) The velocity of the stone at $t=5.00 \text{ s}$.

Solution

Sketch



- (a) Taking the upward direction to be negative and downward to be positive and noticing that at maximum height $v_f = 0$, we get:

$$v_f = v_i + at$$

$$-20.0 \text{ m/s} + 9.8 \text{ m/s}^2 t = 0$$

$$t = \frac{20.0}{9.8} \text{ s} = 2.04 \text{ s}$$

(b) To find the maximum height we use:

$$y = v_i t + \frac{1}{2} a t^2$$

$$y_{\max} = \left(-20.0 \frac{\text{m}}{\text{s}}\right)(2.04 \text{ s}) + \frac{1}{2} \left(\frac{9.80 \text{ m}}{\text{s}^2}\right)((2.04 \text{ s})^2) = -20.4 \text{ m}$$

(c) When the stone is back, $\Delta y = 0$, then,

$$\Delta y = -v_i t + \frac{1}{2} g t^2$$

$$0 = -20.0 t + \frac{1}{2} (9.8 \text{ m/s}^2) t^2$$

Solving for t , we get $t = 0$ or $t = 4.08 \text{ s}$

$t = 0$ corresponds to the time at which the stone starts its motion. $t = 4.08$ is the time taken for the ball to return to the level of the thrower.

(d) The velocity of the ball as it reaches back to the level of the thrower:

$$v_f = v_i + at$$

$$v_f = -20.0 \text{ m/s} + 9.8 \text{ m/s}^2 (4.08 \text{ s}) = +20.0 \text{ m/s}$$

Notice that the velocity of the stone when it arrives back at its original height is equal in magnitude to the stone's initial velocity but opposite in direction.

(e) To get the velocity and position of the stone at $t=5.00 \text{ s}$

$$v_f = v_i + at$$

Taking the upward motion to be negative and downward motion to be positive, we have:

$$v_f = -20.0 \text{ m/s} + (9.8 \text{ m/s}^2)(5.00 \text{ s}) = 29.0 \text{ m/s}$$

We now find the position by

$$\Delta y = (-20.0 \text{ m/s})(5 \text{ s}) + \frac{1}{2} (9.80 \text{ m/s}^2)(5.00 \text{ s})^2 = 22.5 \text{ m}$$

Unit 4

Motion in two dimensions

4.3.1 Displacement, Velocity and Acceleration in Two dimensions

Consider an object moving in space as shown in Figure 3.1.

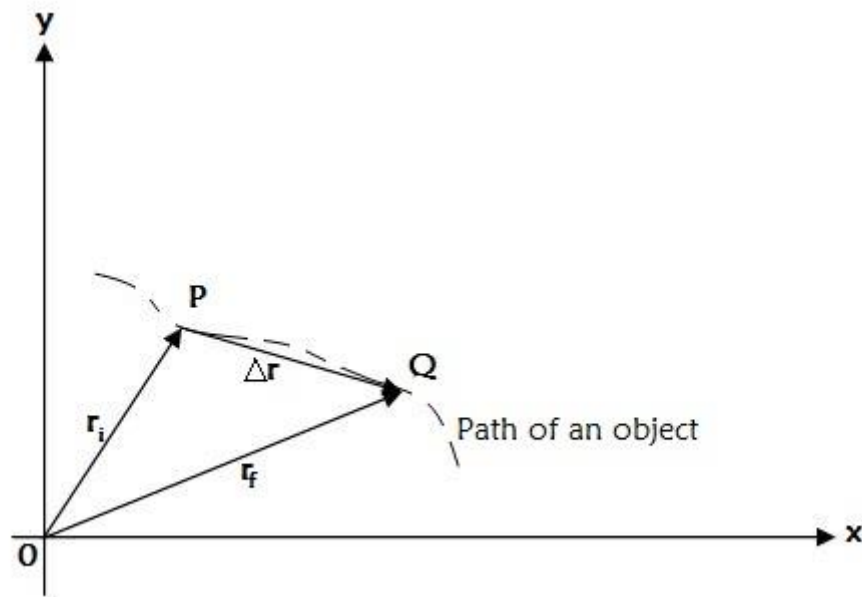


Figure 4.1 Displacement of an object in two dimensions

The object moves from initial point P at time t_i to point Q at time t_f then:

An object's displacement in two dimensions is defined as the change in its position vector given by:

$$\Delta \mathbf{r} = \mathbf{r}_f - \mathbf{r}_i \quad (4.01)$$

Its SI unit: meter (m)

We can now present the generalisations of the definitions of velocity and acceleration given in unit 3 as follows:

An object's average velocity during time interval Δt is its displacement divide by Δt :

$$\mathbf{v}_{av} = \frac{\Delta \mathbf{r}}{\Delta t} \quad (4.02)$$

Its SI unit: meter per second (m/s)

An object's instantaneous velocity is the limit of its average velocity as Δt goes to zero:

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta \mathbf{r}}{\Delta t} \quad (4.03)$$

Its SI unit: meter per second squared (m/s)

The direction of the instantaneous velocity is along the line that is tangent to the object's path and in the direction of motion.

An object's average acceleration is the change in its velocity divide by time interval:

$$\mathbf{a}_{av} = \frac{\Delta \mathbf{v}}{\Delta t} \quad (4.04)$$

SI unit: meter per second squared (m/s^2)

An object's instantaneous acceleration is the limit of its average acceleration as time interval goes to zero:

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta \mathbf{r}}{\Delta t} \quad (4.05)$$

SI unit: meter per second squared (m/s^2)

4.3.2 Projectile motion

Projectile motion is a good example of motion of an object moving in two dimensions. A projectile moves in both x-and y-directions simultaneously.

A body thrown with some initial velocity and then allowed to move under the action of gravity alone is known as a projectile.

The projectile moves in a path, which can be considered as a part of parabola. Such a motion is known as **projectile motion**.

Some examples of projectiles are:

- (i) a bomb thrown from an aero plane
- (ii) a javelin or a shot-put thrown by an athlete
- (iii) motion of a ball hit by a cricket bat

The different types of projectiles are shown in Fig. 4.01.

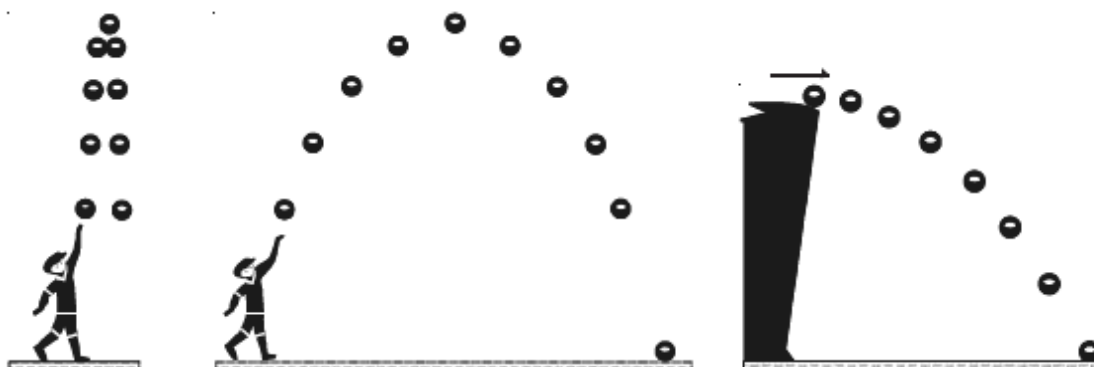


Figure 4.01 Types of projectile motion

A body can be projected in two ways:

- i. It can be projected horizontally from a certain height.
- ii. It can be thrown from the ground in a direction inclined to it.

The projectiles undergo a vertical motion as well as horizontal motion. The two components of the projectile motion are

- i. vertical component and
- ii. Horizontal component

These two components of projectile are perpendicular to each other and are independent of each other

4.3.2.1 Parts of a Projectile motion:

- i. **Angle of projection**
The angle between the initial direction of projection and the horizontal direction through the point of projection is called the angle of projection.
- ii. **Velocity of projection**
The velocity with which the body is projected is known as velocity of projection.
- iii. **Range**
Range of a projectile is the horizontal distance between the point of projection and the point where the projectile hits the ground.
- iv. **Trajectory**
The path described by the projectile is called the trajectory.
- v. **Time of flight**
Time of flight is the total time taken by the projectile from the instant of projection till it strikes the ground

4.3.2.2 Motion of a projectile projected at an angle with the horizontal (oblique projection)

Consider a body projected from a point O on the surface of the Earth with an initial velocity u at an angle θ with the horizontal as shown in Fig. 4.02
The velocity u can be resolved into two components

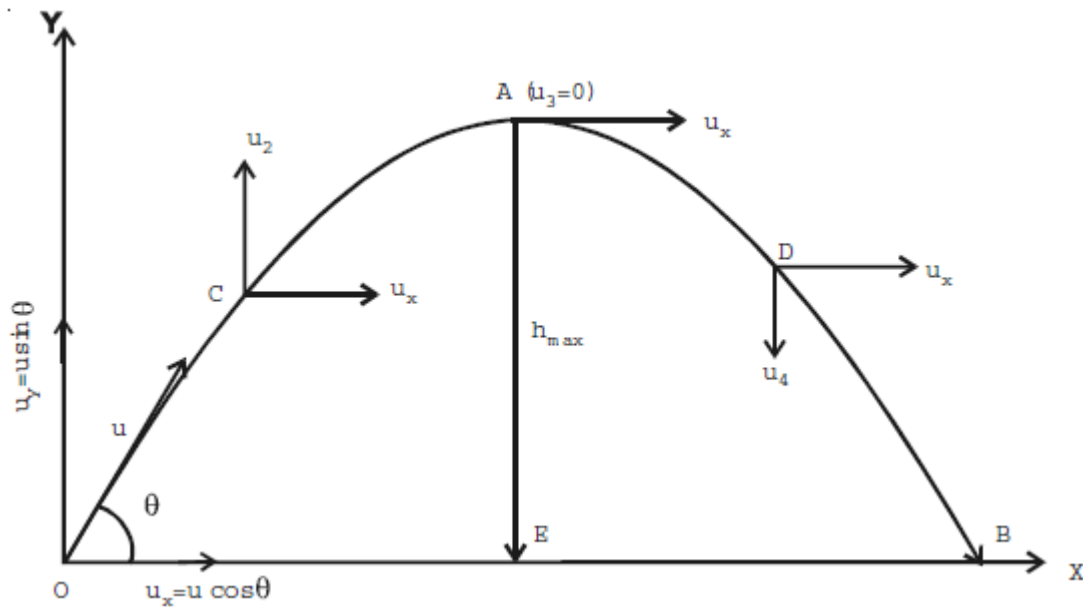


Figure 4.02

(i) $u_x = u \cos \theta$, along the horizontal direction OX and

(ii) $u_y = u \sin \theta$, along the vertical direction OY

The horizontal velocity u_x of the object shall remain constant as no acceleration is acting in the horizontal direction. But the vertical component u_y of the object continuously decreases due to the effect of the gravity and it becomes zero when the body is at the highest point of its path. After this, the vertical component u_y is directed downwards and increases with time till the body strikes the ground at B.

Path of the projectile:

Let t_1 be the time taken by the projectile to reach the point C from the instant of projection.

Horizontal distance travelled by the projectile in time t_1 is,

$$x = u_x t$$

$$x = (u \cos \theta) t \rightarrow t = \frac{x}{u \cos \theta} \quad (4.06)$$

The vertical distance traveled by the projectile, y at time t, is given by

$$y = u_y t - \frac{1}{2} g t^2$$

$$y = (u \sin \theta) t - \frac{1}{2} g t^2 \quad (4.07)$$

Substituting equation 3.6 into equation 3.7 for time t, you have:

$$y = (u \sin \theta) t - \frac{1}{2} g t^2$$

$$y = (u \sin \theta) \left(\frac{x}{u \cos \theta} \right) - \frac{1}{2} g \left(\frac{x}{u \cos \theta} \right)^2$$

$$y = (\tan \theta) x - \frac{g}{2(u \cos \theta)^2} x^2 \quad (4.08)$$

Equation 4.3.8 describes the path taken by a projectile as parabolic (quadratic) in nature which can be rewritten as:

$y = ax^2 + bx + c$ where a , b and c are constants such that $a = -\frac{g}{2(ucos\theta)^2}$, $b = tan\theta$ and $c = 0$

Resultant velocity of the projectile at any instant t

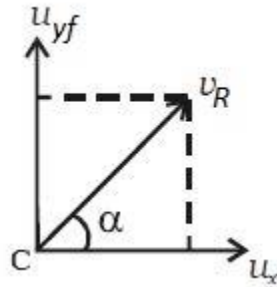


Figure 4.03

At any time, t , $u_x = ucos\theta$ and from the equation of motion $u_{yf} = usin\theta - gt$. The resultant is found by applying Pythagoras theorem:

$$u_R = \sqrt{u_x^2 + u_{yf}^2}$$

$$u_R = \sqrt{(ucos\alpha)^2 + (usina - gt)^2} \quad (4.09)$$

The direction of the resultant velocity can be found by:

$$tana = \frac{u_{yf}}{u_x}$$

$$\alpha = tan^{-1}((usina - gt) / (ucosa)) \quad (4.10)$$

where α is the angle made by u_R with the horizontal line as shown in Figure 3.6

Maximum height reached by the projectile

The maximum vertical displacement produced by the projectile is known as the maximum height reached by the projectile. In Figure 4.02, EA is the maximum height attained by the projectile. It is represented as h_{max} .

At O, $u_{yi} = usin\theta$ and $u_{yf} = 0$

Using equation 1.3.5 which is $u_{yf}^2 = u_{yi}^2 - 2gs$ substituting appropriate variables, you get $0 = (usin\theta)^2 - 2gh_{max}$ and making h_{max} the subject of the formular, you finally get the maximum height of projectile as:

$$h_{max} = \frac{u^2 cos^2 \theta}{2g} \quad (4.11)$$

Time taken to attain maximum height (t_{max})

You are now familiar with the fact that at O, $u_{yi} = usin\theta$ and $u_{yf} = 0$

Applying the equation of motion ($u_{yf} = u_{yi} - gt$) with proper substitutions, you get:

$$0 = u \sin \theta - gt \rightarrow t_{\max} = \frac{u \sin \theta}{g}$$

$\therefore$ the time taken for projectile to reach maximum height is given by:

$$t_{\max} = \frac{u \sin \theta}{g} \quad (4.12)$$

You may also find $h_{\max}$ by substituting $t_{\max}$ into equation $y = u_{yi}t - 1/2 gt^2$. Do it as an activity.

Time taken by the projectile for the whole flight (t_{ft})

You want to find time taken for the projectile to move from O to B in Figure 4.7

At point B, the projectile has returned to where it started. Therefore, its displacement at point B is zero. Applying equation $y = u_{yi}t - 1/2 gt^2$ and making substitutions, you get:

$0 = (u \sin \theta)t_{ft} - 1/2 gt_{ft}^2$ and solving for t_{ft} , you get time for the whole flight as:

$$t_{ft} = \frac{2u \sin \theta}{g} \quad (4.13)$$

Note that, the time taken by the projectile for the whole flight is just twice the time for the projectile to reach maximum height.

Horizontal range (R)

From Figure 4.02, Range R is the distance between OB.

Along OB, the projectile moves with constant velocity i.e it has zero acceleration.

Therefore:

$R = u_x t = (u \cos \theta)t$, t is time for the whole flight.

$$R = (u \cos \theta)t_{ft} = (u \cos \theta)\left(\frac{2u \sin \theta}{g}\right)$$

$$R = \frac{2u^2 \cos \theta \sin \theta}{g}$$

$$R = \frac{u^2 \sin 2\theta}{g} \quad (4.14)$$

Maximum Range

From equation 4.3.13, The Range R is maximum only if $\sin 2\theta$ is maximum

$$i.e \sin 2\theta = 1$$

$$2\theta = \sin^{-1}(1)$$

$$2\theta = 90^\circ$$

$$\therefore \theta = 45^\circ$$

Then

$$R_{max} = \frac{u^2 \sin(2 \times 45^\circ)}{g} = \frac{u^2 \times 1}{g} = \frac{u^2}{g}$$

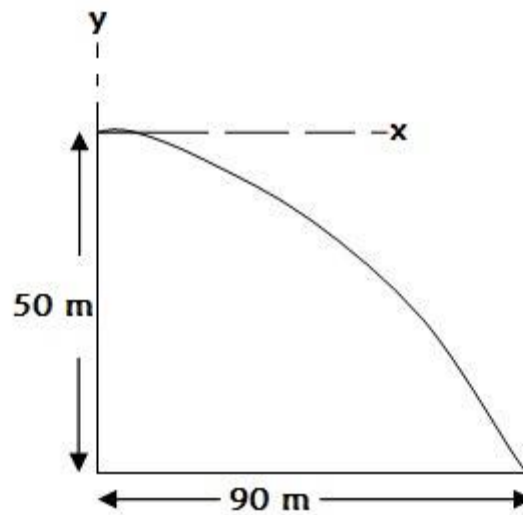
$$\therefore R_{max} = \frac{u^2}{g} \quad (4.15)$$

Example 4.1

A movie student driver on a motorcycle speeds horizontally off a 50 m high cliff. How fast must the motorcycle leave the cliff-top if it is to land on level ground below, 90.0 m from the base of the cliff where the cameras are?

Solution

Sketch diagram



Taking the downward motion to be positive, we have $v_0 = ?$, $y = 50 \text{ m}$, $g = 9.8 \text{ m/s}^2$
 $x = 90 \text{ m}$, and $v_{0y} = 0$ since it speeds horizontally,

$$y = v_{0y}t + \frac{1}{2}gt^2$$

$$50 = \frac{1}{2}(9.8)t^2$$

Solving for t , we get $t = 3.19 \text{ s}$.

To get the initial velocity, v_{0x} , we apply:

$$x = v_{0x}t$$

$$90 = v_{0x}(3.19)$$

We solve for v_{0x} , and get:

$$v_{0x} = 28.2 \text{ m/s}$$

Example 4.2

A football is kicked at an angle $\theta = 37.0^\circ$ with velocity of 20 m/s . Calculate the maximum height reached by the ball.

Solution

At maximum height $v_{0y} = 0$, and this occurs at time

$$t = \frac{v_{0y}(\sin \theta)}{g}$$

$$t = \frac{20.0 \sin 37.0^\circ}{9.8} = 1.22 \text{ s}$$

We then find the maximum height:

$$y_{max} = v_{0y}t + \frac{1}{2}gt^2$$

And taking the upward to be negative and the downward to be positive, we have:

$$y_{max} = (-20 \sin 37^\circ)(1.22 \text{ s}) + \frac{1}{2}\left(9.8 \frac{\text{m}}{\text{s}^2}\right)(1.22 \text{ s})^2 = 7.35 \text{ m}$$

4.3.3 Relative motion

Relative motion is about relating the measurements of two different observers, one moving with respect to the other. Measurements of velocities depend on the reference frame of the observer. Recall that reference frames are just the coordinate systems from which motion measurements are based. Most of the time we use a stationary frame of reference relative to the Earth, but occasionally we use a moving frame of reference associated with a bus, car or a plane moving with constant velocity relative to the Earth.

In two dimensions, let E be an observer, assumed stationary with respect to the Earth.

Let two cars labeled A and B, and introduce the following notation see Figure 4.02.

r_{AE} = the position of Car A as measured by E (in a coordinate system fixed with respect to the Earth)

r_{BE} = the position of Car B as measured by E

r_{AB} = the position of Car A as measured by an observer in Car B

Then:

$$\mathbf{r}_{AB} = \mathbf{r}_{AE} - \mathbf{r}_{BE} \quad (4.16)$$

And the rate of change of these quantities with time gives us relative velocities as:

$$\mathbf{v}_{AB} = \mathbf{v}_{AE} - \mathbf{v}_{BE} \quad (4.17)$$

For any two objects or reference frames, A and B, the velocity of A relative to B has the same magnitude, but opposite direction, as the velocity of B relative to A:

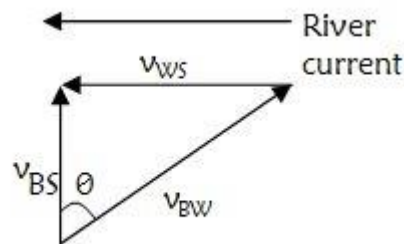
$$v_{BA} = -v_{AB} \quad (4.17)$$

Example 4.1

A boat's speed in still water is $v_{BW} = 2.00 \text{ m/s}$. If the boat is to travel directly across a river whose current has speed of $v_{WS} = 1.50 \text{ m/s}$, at what upstream angle must the boat head?

Solution

Letting v_{BS} be the velocity of boat as seen from the shore, the situation can be presented diagrammatically as follows:



$$\sin \theta = \frac{v_{WS}}{v_{BW}} = \frac{1.50 \text{ m/s}}{2.00 \text{ m/s}}$$

$$\theta = \sin^{-1} \left(\frac{1.50}{2.00} \right)$$

$$\therefore \theta = 48.59^\circ$$

So the boat must head upstream at 48.59° angle.

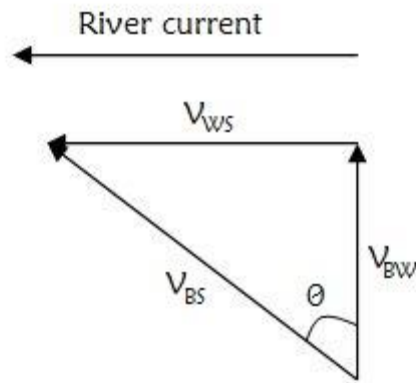
Example 4.2.o

The same boat in example 4.1, ($v_{BW} = 2.00 \text{ m/s}$) now heads directly across the stream whose current is still 1.5 m/s .

- What is the velocity of (magnitude and direction) of the boat relative to the shore?
- If the river is 110 m wide, how long will it take to cross and how far downstream will the boat be then?

Solution

-



$$\begin{aligned} \mathbf{v}_{BS} &= \mathbf{v}_{BW} + \mathbf{v}_{WS} \\ v_{BS} &= \sqrt{v_{BW}^2 + v_{WS}^2} = \sqrt{(2.00 \text{ m/s})^2 + (1.50 \text{ m/s})^2} \\ v_{BS} &= 2.5 \text{ m/s} \end{aligned}$$

(b) Given the river's width $D = 110 \text{ m}$, and using the definition of velocity, we solve for

$$\begin{aligned} t &= \frac{D}{v_{BW}} \\ t &= \frac{110 \text{ m}}{2.00 \text{ m/s}} = 55 \text{ s} \end{aligned}$$

The boat will have been carried downstream in this time a distance of:

$$d = v_{WS}t = (1.50 \text{ m/s})(55 \text{ s}) = 82.5 \text{ m}$$

Unit 5

Forces and Motion

5.3 Force and Motion

5.3.1 What is force?

A force is anything that can cause a change to an object. Forces can:

- Change the shape of an object

- Move or stop an object
- Change the direction of a moving object
- Produce heat in an object

A force can be contact or noncontact

A contact force is in contact with an object to cause a change e.g

- The force that is used to push or pull things, like on a door to open or close it
- The force that a sculptor uses to turn clay into a pot
- The force of the wind to turn the windmill

A non-contact force (Field force) does not have to touch an object to cause a change.
E.g

- Gravitational force, like the earth pulling the moon towards itself
- Electric force, like a proton and an electron attracting each other
- Magnetic force, like a magnet pulling a paper clip towards itself

The unit of force is Newton (N). 1 Newton is a force that produces an acceleration of 1ms^{-2} when applied on a mass of 1 kg. i.e $1\text{N} = 1\text{kgms}^{-2}$. Force is a vector physical quantity i.e it is completely described by both magnitude and direction.

Resultant force or net force is a single force representing the vector sum of the forces that act on the object.

Activity 5.1

1. Find the resultant force of the two forces $F_1 = 3\text{ N}$ and $F_2 = 4\text{ N}$ which are at an angle of 60° to each other.

5.3.2 Newton's laws of motion

5.3.2.1 Newton's first law of motion

It states that an object will remain at its state of rest or state of motion with constant velocity except when it is acted upon by a non-zero net force. This means that force is not needed to maintain its motion. According to this law, an object may either be in a static or dynamic equilibrium.

Newton's first law is also called the law of inertia. Inertia is the resistance (inability) of an object to start moving or to stop moving once it started.

This resistance is directly proportional to the mass of object.

There are three types of inertia:

(i) Inertia of rest

It is the inability of the body to change its state of rest by itself.

Examples

(i) A person standing in a bus falls backward when the bus suddenly starts moving.

This is because, the person who is initially at rest continues to be at rest even after the bus has started moving.

(ii) A book lying on the table will remain at rest, until it is moved by some external agencies.

(iii) When a carpet is beaten by a stick, the dust particles fall off vertically downwards once they are released and do not move along the carpet and fall off.

(ii) Inertia of motion

Inertia of motion is the inability of the body to change its state of motion by itself.

Examples

(a) When a passenger gets down from a moving bus, he falls down in the direction of the motion of the bus.

(b) A passenger sitting in a moving car falls forward, when the car stops suddenly.

(c) An athlete running in a race will continue to run even after reaching the finishing point.

(iii) Inertia of direction

It is the inability of the body to change its direction of motion by itself.

Examples

When a bus moving along a straight line takes a turn to the right, the passengers are thrown towards left. This is due to inertia which makes the passengers travel along the same straight line, even though the bus has turned towards the right.

From the first law, you can infer that to change the state of rest or uniform motion, an external agency called force is required.

Therefore, force can also be defined as an agency which when acting on a body changes or tends to change the state of rest or of uniform motion of the body along a straight line.

Mass and Inertia

You recall that **Inertia** is the tendency of an object to resist being accelerated by a force and that inertia depends on mass.

Therefore mass, m , can be defined as a measure of an object's inertia

Mass can also be defined as a measure of the quantity of substance in a physical object

The SI unit of mass is kilogram (kg).

Mass is a scalar quantity

Mass is not weight; mass is a property of an object, whereas weight is the force exerted on the object by gravity.

5.3.2.2 Newton's Second Law of Motion and Weight

It can be stated as acceleration is equal to force divided by mass

Expressed by the equation:

$$\mathbf{F_R} = m\mathbf{a} \quad (5.04)$$

Equation 5.04 can also be written as:

$$\sum \mathbf{F} = m\mathbf{a} \quad (5.05)$$

Writing equation 5.05 in Vector components you get:

$$\sum F_x = m\mathbf{a}_x \quad (5.06)$$

$$\sum F_y = m\mathbf{a}_y \quad (5.07)$$

From Newton's Second Law of motion, the magnitude of gravitational force (weight) can be found by:

$$\mathbf{F_g} = m\mathbf{g} \quad (5.08)$$

Where F_g is gravitational force, m is the mass of an object and g is the object's acceleration due to gravity.

5.3.2.3 Newton's Third Law of Motion

This law is the action-reaction law. It states that

"To every action there is always an equal and opposite reaction"

OR

"If object A exerts a force on object B then object B exerts a force on object A that is equal in magnitude but opposite in direction"

Newton's third law of motion implies that if an object A exerts force on object B, then object B will exert an equal amount of force on object A but in opposite direction. These forces do not cancel each other because they act on different objects. The effect of third law of motion can be observed in many activities in our everyday life. Some examples are:

- (i) When a bullet is fired from a gun with a certain force (action), there is an equal and opposite force exerted on the gun in the backward direction (reaction).
- (ii) When a man jumps from a boat to the shore, the boat moves away from him. The force he exerts on the boat (action) is responsible for its motion and his motion to the shore is due to the force of reaction exerted by the boat on him.
- (iv) The swimmer pushes the water in the backward direction with a certain force (action) and the water pushes the swimmer in the forward direction with an equal and opposite force (reaction).

5.3.3 Application of Newton's Laws of motion

The following steps can be followed when solving problems related to Newton's laws of motion:

1. read the problem carefully at least once
2. draw a picture of the system, identify the object of primary interest, and indicate forces with arrows
3. label each force in the picture in the way that will bring to mind what physical quantity the label stands for e.g T is for tension
4. Draw a free body diagram of the object of interest, based on the labeled picture. If additional objects are involved, draw separate free-body diagrams for them. Choose convenient coordinates for for each object
5. Apply Newton's second law. The x-and y-components of Newton's second law should be taken from the vector equation and written individually. This usually result in two equations and two unknowns
6. solve for the desired unknown quantity, and substitute the numbers

5.3.3.1 Objects in translational Equilibrium

Objects that are at rest or are moving with constant velocity are said to be in equilibrium. In this situation acceleration a is zero and Newton's second law applied to an object in translational equilibrium gives:

$$\sum \mathbf{F} = 0 \quad (5.09)$$

This equations tells you that the vector sum of all forces (net force) acting on an object in equilibrium is equal to zero.

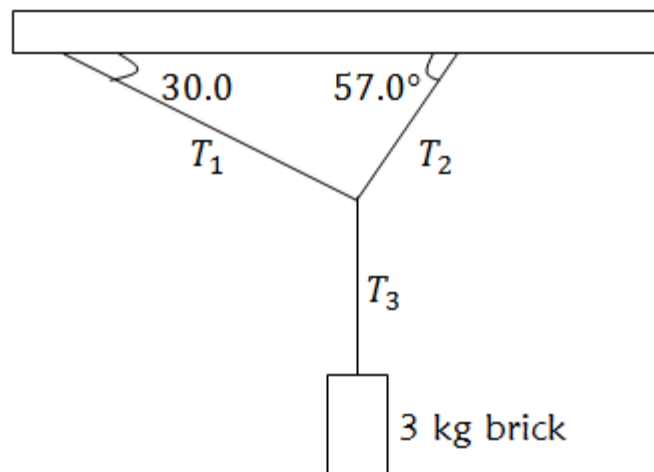
Equation 5.09 can be written in the following equivalent vector forms in two dimensions:

$$\Sigma F_x = 0 \text{ and } \Sigma F_y = 0 \quad (5.10)$$

Note that an object at rest and an object moving with constant velocity are both at translational equilibrium because in both cases acceleration is zero giving us a net force equal to zero.

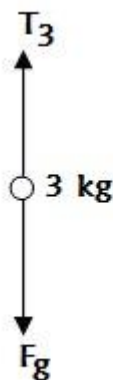
Example

A brick of mass 3 kg hangs from a vertical cable tied to two other cables that are fastened to a support as shown in the Figure below. Find the tension in each of the three cables.

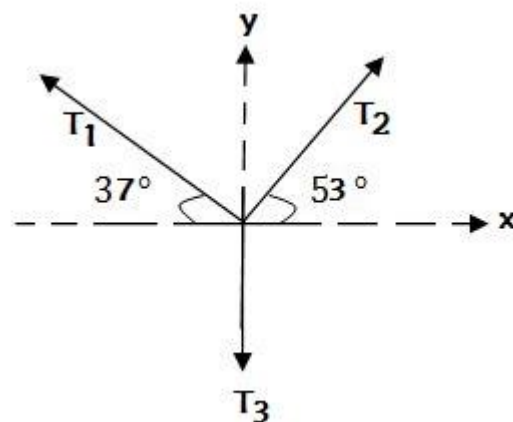


Solution

We draw the free-fall diagrams:



(a)



(b)

From the first free-fall diagram (a) we write the following equation:

$$\sum F_y = 0 \rightarrow T_3 - F_g = 0$$

$$T_3 = F_g = mg = 3kg(9.8 \text{ m/s}^2)$$

$$T_3 = 29.4 \text{ N}$$

We write the two equations from the second free-fall diagram (b):

$$\sum F_x = 0 \rightarrow T_2 \cos 53^\circ - T_1 \cos 37^\circ = 0 \quad (i)$$

$$\sum F_y = 0 \rightarrow T_2 \sin 53^\circ + T_1 \sin 37^\circ - T_3 = 0 \quad (ii)$$

From Equation I, $T_2 = T_1 \left(\frac{\cos 37^\circ}{\cos 53^\circ} \right) = 1.327 T_1$

Substitute this result into ii,

$$1.327 T_1 \sin 53^\circ + T_1 \sin 37^\circ - T_3 = 0$$

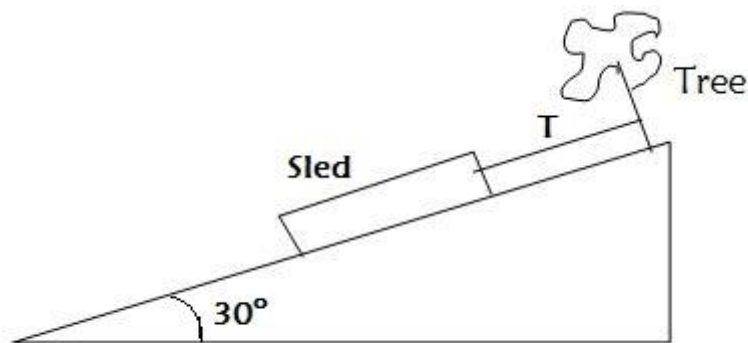
$$T_1 = \frac{T_3}{1.327 \sin 53^\circ + \sin 37^\circ} = \frac{29.4}{1.662} = 17.690 \text{ N}$$

$$T_2 \cos 53^\circ = T_1 \cos 37^\circ \rightarrow T_2 = \frac{T_1 \cos 37^\circ}{\cos 53^\circ}$$

$$= \frac{17.690 \cos 37^\circ}{\cos 53^\circ} = 23.475 \text{ N}$$

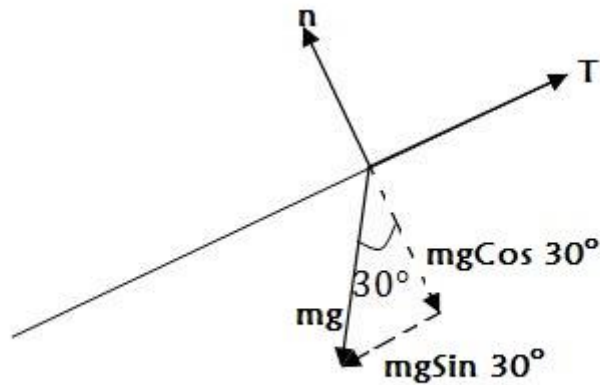
Example

A sled is tied to a tree on a frictionless, snow-covered hill as shown in the Figure below. If the sled weighs 77.0 N, find the magnitude of the tension T exerted by the rope on the sled and the normal force n exerted by the hill on the sled.



Solution

Draw a free-fall diagram



$$n - mg \cos 30^\circ = 0$$

$$n = mg \cos 30^\circ = 77.0 \cos 30^\circ = 66.68 \text{ N}$$

$$T + f - mg \sin 30^\circ$$

$$T + 0 - 77.0 \sin 30^\circ$$

$$T = 77.0 \sin 30^\circ = 38.5 \text{ N}$$

$\therefore$ The tension $T = 66.68 \text{ N}$ and normal force $n = 38.50 \text{ N}$

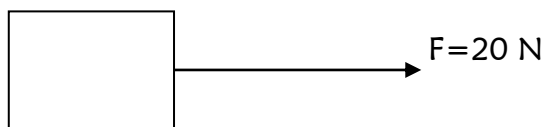
5.3.3.2 Accelerating objects and Newton's second law

When a net force acts on an object, the object accelerates and Newton's second Law is applied.

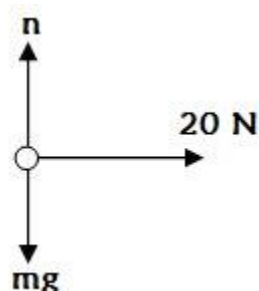
Example

The weight of a box containing books is 300 N. If a student pulls on the rope with a constant force of 20 N, what is the acceleration of the crate and how far will it move in 2.00 s? Assume the box starts from the rest and that there is no friction between the surface and the box.

Solution



And its free-fall diagram is



There is no motion in y direction and therefore no acceleration in y direction

We find mass from $mg = 300\text{ N} \rightarrow m = \frac{300}{9.8} = 30.6\text{ kg}$

$$ma_x = F$$

$$a_x = \frac{20\text{ N}}{30.6\text{ kg}} = 0.654\text{ m/s}^2$$

We now apply the following equation of motion:

$$x_f - x_i = v_i t + \frac{1}{2}at^2$$

$$x_f - 0 = (0)(2.00\text{ s}) + \frac{1}{2}(0.654)(2.00\text{ s})^2 = 1.308\text{ m}$$

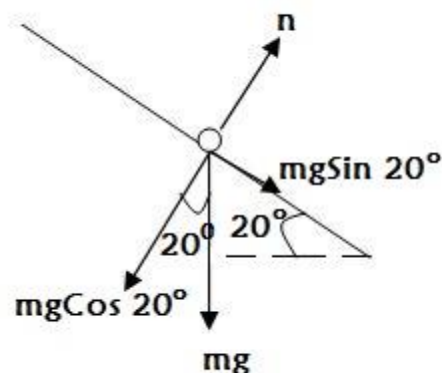
Example

A toy car of mass 0.5 kg is on a frictionless inclined plane at an angle of 20° .

- (a) Determine its acceleration down the incline plane.
- (b) If the length of the incline plane is 5.0 m and the car starts from rest at the top, how long does it take to travel to the bottom?
- (c) What is the car's speed at the bottom?

Solution

- (a) Free-fall diagram:



$$mg \sin 20^\circ = ma_x$$

$$a_x = \frac{mg \sin 20^\circ}{m} = g \sin 20^\circ = \left(\frac{9.8\text{ m}}{\text{s}^2}\right) \sin 20^\circ = 3.352\text{ m/s}^2$$

(b) $\Delta x = v_i t + \frac{1}{2}at^2$

$$5.0 = (0)t + \frac{1}{2}(3.352)t^2$$

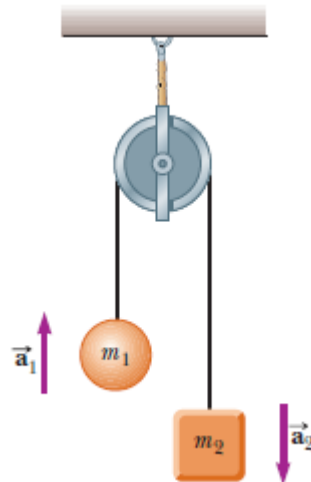
$$t = \sqrt{\frac{5.0 \times 2}{3.352}} = 1.727\text{ s}$$

(c) $v_f = v_i + at$

$$v_f = 0 + (3.352 \text{ m/s}^2)(1.727 \text{ s}) = 5.789 \text{ m/s}$$

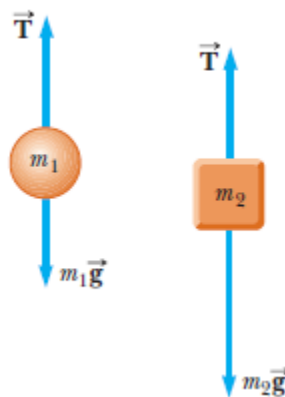
Example

Two objects of mass m_1 and m_2 , with $m_2 > m_1$, are connected by a light, inextensible cord and hung over a frictionless pulley, as in the Figure below. Both cord and pulley have negligible mass. Find the magnitude of the acceleration of the system and the tension in the cord.



Solution

the following are the free-fall diagrams for m_1 and m_2 :



For m_1 , we write the following Equation:

$$T - m_1g = m_1a_y \quad (i)$$

For m_2 , we have the following Equation:

$$m_2g - T = m_2a_y \quad (ii)$$

Adding the two equations, you get:

$$(m_2 - m_1)g = (m_2 + m_1)a_y$$

$$a_y = \frac{(m_2 - m_1)g}{(m_2 + m_1)}$$

Lets use Equation I to get T:

$T = m_1 a_y + m_1 g$ and Substituting a_y , we get:

$$T = m_1 \left(\frac{(m_2 - m_1)g}{(m_2 + m_1)} \right) + m_1 g$$

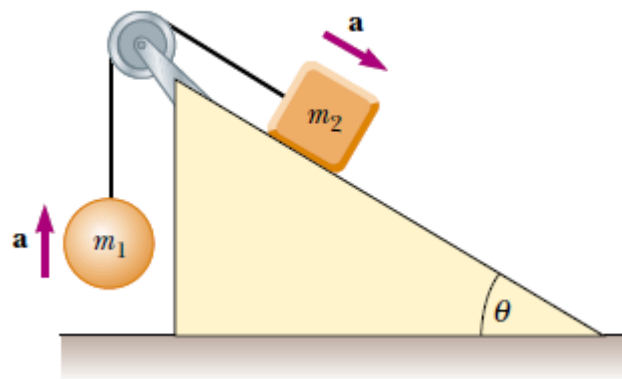
After further simplification, we get:

$$T = \frac{2m_1 m_2 g}{(m_2 + m_1)}$$

Note that the acceleration of the second block is the same as that of the first, but negative. When m_2 gets very large compared with m_1 , the acceleration of the system approaches g , as expected, because m_2 is falling nearly freely under the influence of gravity. Indeed, m_2 is only slightly restrained by the much lighter m_1 .

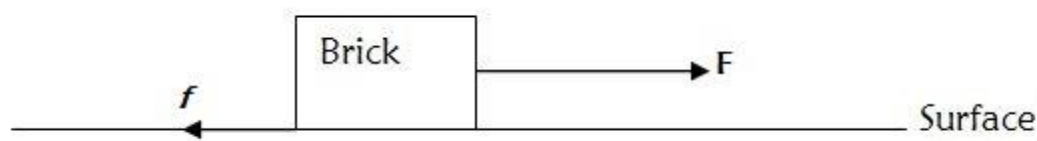
Activity 5.1

1. Why does a child in a wagon seem to fall backward when you give the wagon a sharp pull?
2. A ball of mass m_1 and a block of mass m_2 are attached by a lightweight cord that passes over a frictionless pulley of negligible mass, as in Figure below. The block lies on a frictionless incline of angle θ . Find the magnitude of the acceleration of the two objects and the tension in the cord.

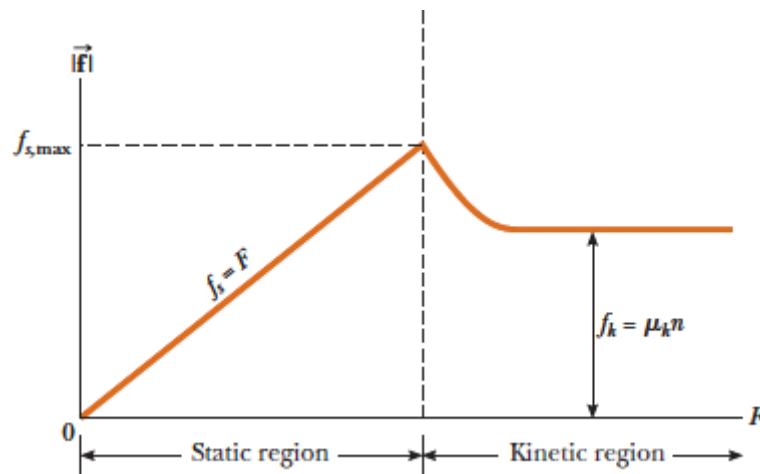


5.3.4 Friction Force

Consider a applied force F on a brick at rest on the surface as shown in the Figure below:



The applied force F is opposed by friction force, f . Before the brick starts moving F is equal to f and f_s is directly proportional to the applied force F until f_s reaches its maximum value $f_{s,max}$ as shown in the graph of Figure



f_s is maximum when the brick is on the verge of moving. At this point F exceeds $f_{s,max}$, the brick accelerates to the right. When the object is in motion the friction force is less than applied force. The friction force for an object in motion is called kinetic friction, f_k .

The magnitude of maximum static friction is given by:

$$f_{s,max} = \mu_s n$$

where μ_s is constant called coefficient of static friction. And n is the magnitude of the normal force exerted by one surface on the other.

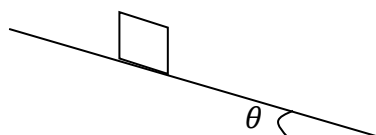
The magnitude of kinetic friction is found by:

$$f_k = \mu_k n$$

Where μ_k is constant called the coefficient of kinetic friction.

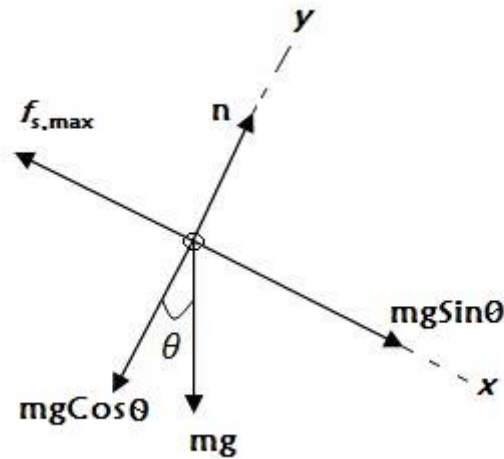
Example

Suppose a block with a mass of 2.50 kg is resting on a ramp. If the coefficient of static friction between the block and ramp is 0.350, what maximum angle can the ramp make with the horizontal before the block starts to slip down?



Solution

Free-fall diagram



$$n = mg \cos \theta$$

$$f_{s,max} = mg \sin \theta \quad (i)$$

And you know that

$$f_{s,max} = \mu_s n = \mu_s (mg \cos \theta) \quad (ii)$$

Equating the two Equations, you have:

$$\mu_s (mg \cos \theta) = mg \sin \theta$$

$$\tan \theta = \mu_s$$

$$\theta = \tan^{-1}(\mu_s) = \tan^{-1}(0.350) = 19.29^\circ$$

Unit 6

Work, Energy and Power

6.3.1 Work done

Work is done when a force $\mathbf{F}$, exerted on an object causes the object to move in the direction of the force. Work is a scalar quantity.

If a constant force $\mathbf{F}$ acts on an object which moves a distance Δx in the direction of the force as shown in Figure 6.1, then work done on the object by this force is given by:

$$W = F \Delta x \quad (6.01)$$

Where W = work done in joules (J)

F = Constant applied force in newton (N)

$\Delta x = \text{displacement measured in meters (m)}$

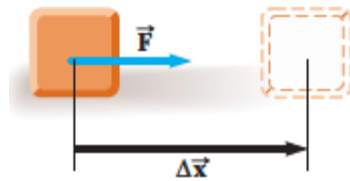


Figure 6.1 A constant force F in the same direction as the displacement Δx , does work $F\Delta x$

If a constant force F acts on an object at an angle which moves a distance Δx in the x -direction as shown in Figure 6.2, then work done on the object by this force is given by:

$$W = (F\cos\theta)\Delta x \quad (6.02)$$

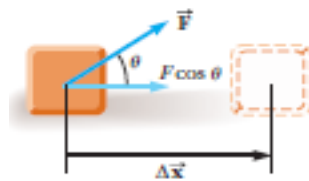


Figure 6.2 A constant force exerted at an angle θ with respect to the displacement, Δx , does work $(F\cos\theta)\Delta x$.

Where:

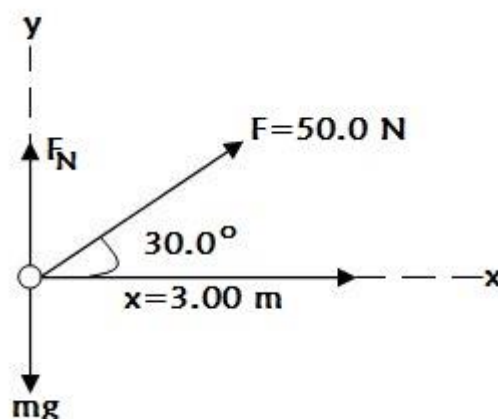
$F = \text{magnitude of the force}$, Δx is the magnitude of the displacement and θ is the angle between the directions of F and Δx .

Example 6.1

A man cleaning his apartment pulls the canister of a vacuum cleaner with a force of magnitude $F = 50.0 \text{ N}$ at an angle of 30.0° with the horizontal. He moves the vacuum cleaner a distance of 3.00 m . calculate the work done by the 50.0 N force.

Solution

We draw the free body diagram as follows:



From the free the free body diagram, we notice that F_N and mg do no work because there is no movement in y-direction. Therefore,

$$W = F_x s = (50.0 \cos 30.0^\circ)(3.00 \text{ m}) = 130 \text{ J}$$

Activity 6.1

1. Define work
2. A 2.0 kg object is moving at 3m/s. A 4.0 N force is applied in the direction of motion and then removed after the object has traveled an additional 5.0 m. calculate the value of work done on the object
3. A man wishes to pull a crate 15 m across a rough floor by exerting a force of 100 N. the coefficient of kinetic friction is 0.25. Find the angle between the force and the horizontal distance for the man to do the least work.

6.3.2 Energy

Energy is the ability to do work. The SI unit of energy is joule (J).

There are many forms of energy. Here the focus is on mechanical energy which consists of Kinetic and Potential energy.

6.3.2.1 Kinetic Energy and the Work-Energy Theorem

Kinetic energy is the energy possessed by an object by virtue of movement.

Suppose an object of mass, m , moves under the action of constant net force, F shown in Figure 6.3.

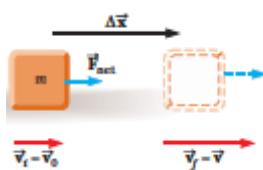


Figure 6.3

The force is constant and therefore the object moves with constant acceleration. If the object is displaced by Δx , then the work done by F is:

$$W_{net} = F_{net} \Delta x = ma \Delta x \quad (6.03)$$

You know that when the object moves with constant acceleration, then the following equation holds:

$$\begin{aligned} v^2 &= u^2 + 2a\Delta x \\ \rightarrow a\Delta x &= \frac{m(v^2 - u^2)}{2} \end{aligned} \quad (6.04)$$

Substituting Equation 6.4 into equation 6.3 for $a\Delta x$, you get:

$$\begin{aligned} W_{net} &= m \left(\frac{v^2 - u^2}{2} \right) \\ W_{net} &= \frac{1}{2}mv^2 - \frac{1}{2}mu^2 \end{aligned} \quad (6.05)$$

The Kinetic Energy (KE) of an object of mass m moving with speed v is

$$KE = \frac{1}{2}mv^2 \quad (6.06)$$

Equation 6.5 is Known as Work-Energy Theorem.

Work-Energy theorem states that the net work done on an object is equal to the change of object's Kinetic energy:

$$W_{net} = KE_f - KE_i = \Delta KE \quad (6.07)$$

The change in kinetic energy is due to the object's change in speed.

Example 6.2

A 1400 kg car has a net forward force of 4500 N applied to it. The car starts from rest and travels down a horizontal high-way. What are its kinetic energy and speed after it has traveled 100 m? (Ignore losses in kinetic energy because of friction and air resistance.)

Solution

$v_i = 0$, and work is given by:

$$\begin{aligned} W_{net} &= KE_f - KE_i = \Delta KE \\ &= \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 \\ &= \frac{1}{2}(1400)v_f^2 - \frac{1}{2}(1400)(0)^2 = 700v_f^2 \end{aligned} \quad (i)$$

And the work done by the net force on the car is:

$$W_{net} = Fs = 4500 \text{ N} \times 100 \text{ m} = 4.50 \times 10^5 \text{ J} \quad (ii)$$

To get the final speed, we equate the two equations:

$$\begin{aligned} 700v_f^2 &= 4.50 \times 10^5 \text{ J} \\ v_f &= \sqrt{\frac{4.50 \times 10^5}{700}} = 25.4 \text{ m/s} \end{aligned}$$

6.3.3 Conservative and non-conservative forces

A force is said to be conservative if the work it does in moving an object between two points is the same no matter what path is taken, otherwise it is non-conservative. Force of gravity is the best example of conservative force.

Non-conservative force disperses object's energy which is in the form of heat or sound energy. Examples of non-conservative forces are kinetic friction and air drag.

Suppose W_{nc} be work done by non – conservative force and W_c be work done by conservative force, the net work done is given by the sum of W_{nc} and W_c i.e

$$W_{net} = W_{nc} + W_c$$

From equation 5.7 $W_{net} = \Delta KE$, you get:

$$W_{nc} + W_c = \Delta KE \quad (6.08)$$

6.3.4 Potential Energy

Potential energy is the energy possessed by an object by virtue of its position.

6.3.4.1 Gravitational Work and Potential Energy

Gravity is a conservative force and for every conservative force potential energy can be found.

Consider a book of mass m falls from a height h_i to a height h_f and assume that the only force acting on the book is gravitation. The magnitude of gravitational force F_g is mg and the displacement Δh is $h_i - h_f$. The angle between F_g and Δh is 0° since both point down as shown in Figure 6.5

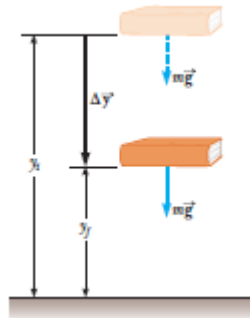


Figure 4.5

Applying the definition of work:

$$W_g = (F_g \cos \theta) \Delta h = mg(h_i - h_f) \cos 0^\circ = -mg(h_f - h_i) \quad (6.09)$$

Suppose that the system involves only conservative and non conservative forces and you know that W_g is work done by conservative force, then:

$$W_{nc} + W_g = \Delta KE$$

and

$$W_{nc} - mg(h_f - h_i) = \Delta KE \quad (6.10)$$

Making W_{nc} the subject of the formula, you get:

$$\begin{aligned} W_{nc} &= \Delta KE + mg(h_f - h_i) \\ W_{nc} &= \Delta KE + mgh_f - mgh_i \end{aligned} \quad (6.11)$$

The gravitational energy of a system is:

$$PE = mgh \quad (6.12)$$

Where g is the acceleration due to gravity and h is the vertical position of the mass m relative to the surface of the earth.

From equation 6.3.9:

$$\begin{aligned}W_g &= -mg(h_f - h_i) \\ \therefore W_g &= -(mgh_f - mgh_i) \\ W_g &= -(PE_f - PE_i)\end{aligned}\tag{6.13}$$

Equation 6.13 tells you that the work done by gravity is the same as negative change in gravitational potential energy.

Using Equation 6.13 in Equation 6.11 then work-energy theorem is given by:

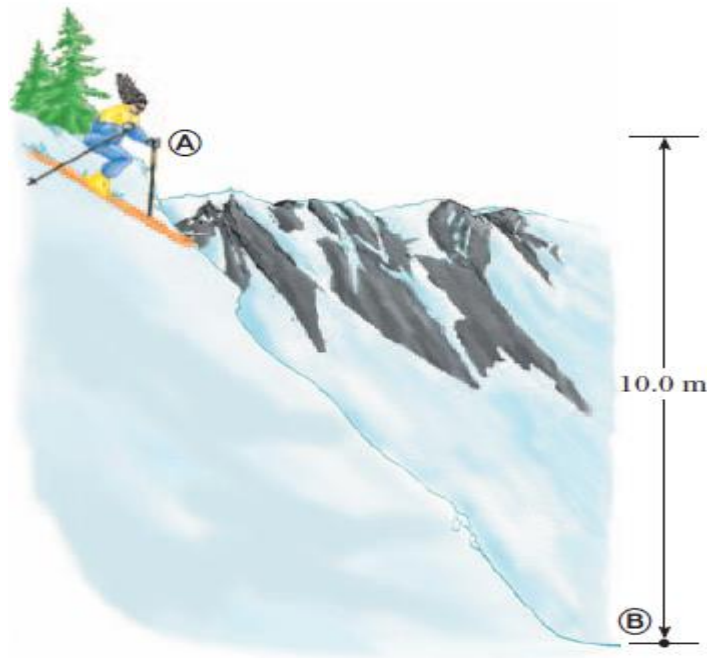
$$W_{nc} = (KE_f - KE_i) + (PE_f - PE_i)\tag{6.14}$$

which states that the work done by non-conservative forces is equal to the change of kinetic energy plus the change of gravitational potential energy.

Example 6.3

A 60.0 kg skier is at the top of a slope. At the initial point A the skier is 10.0 m vertically above point B.

- Setting the zero level for gravitational potential energy at B, find the gravitational potential energy at A and B.
- Repeat this problem with zero level at point A



Solution

-

$$PE_i = mgy_i = 60 \text{ kg} \times 9.80 \times 10.0 \text{ m} = 5880 \text{ J}$$

Because $PE_f = 0$, the difference in potential energy is

$$PE_i - PE_f = 5880 \text{ J} - 0 \text{ J} = 5880 \text{ J}$$

$$\text{b. } PE_f = mgy_f = 60 \text{ kg} \times 9.80 \times (-10.0 \text{ m}) = -5880 \text{ J}$$

In this case the distance $y_f = 0$.

$$PE_i - PE_f = 0 \text{ J} - (-5880 \text{ J}) = 5880 \text{ J}$$

The potential energy at the top of the slope is greater than the potential energy at the bottom by 5880 J regardless of the zero level selected.

Activity 6.3

1. A projectile of mass 0.50 kg is fired with an initial speed of 10 m/s at an angle of 60° above the horizontal. Find the potential energy of the projectile.

6.3.5 Gravity and the Conservation of Mechanical Energy

In the absence of non-conservative forces on the system, $W_{nc} = 0$ and Equation 6.14 becomes:

$$0 = (KE_f - KE_i) + (PE_f - PE_i)$$

Putting initial terms and final terms together, you get:

$$KE_i + PE_i = KE_f + PE_f \quad (6.15)$$

You see from equation 6.15 that the sum of kinetic energy and potential energy remains constant all times i.e mechanical energy is conserved for an isolated system.

Equation 6.15 can be rewritten as:

$$E_{mec} = KE + PE \quad (6.16)$$

where E_{mec} is the total mechanical energy.

6.3.6 The law of conservation of Mechanical energy

It states that in an isolated system in which only conservative force is involved the total mechanical energy of the system remains constant i.e the total initial mechanical energy is equal to total final mechanical energy.

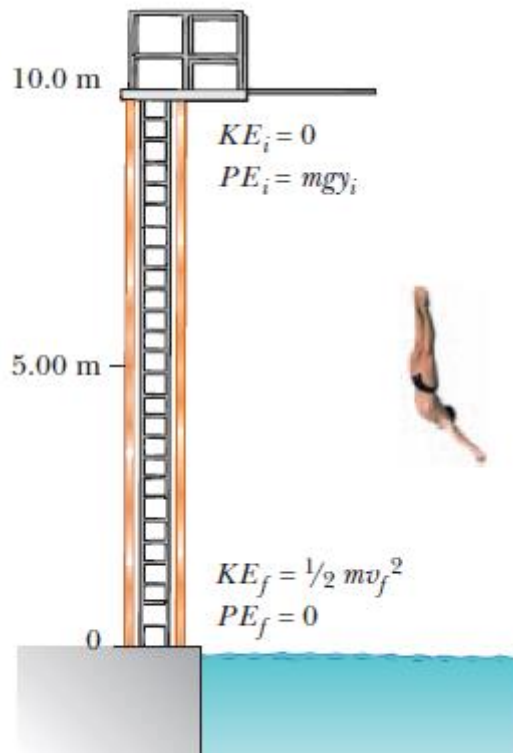
You can write equation 6.3.15 in the form:

$$\frac{1}{2}mu^2 + mgh_i = \frac{1}{2}mv^2 + mgh_f \quad (6.17)$$

if the only force doing work on an isolated system is conservative.

Example 6.4

A diver of mass m drops from a board 10.0 m above the water's surface, as in Figure 6.16. Neglect air resistance. **(a)** Use conservation of mechanical energy to find his speed 5.00 m above the water's surface. **(b)** Find his speed as he hits the water.



Solution

$$\begin{aligned}
 KE_i + PE_i &= KE_f + PE_f \\
 \frac{1}{2}mv_i^2 + mgy_i &= \frac{1}{2}mv_f^2 + mgy_f \\
 0 + gy_i &= \frac{1}{2}v_f^2 + gy_f \\
 v_f &= \sqrt{2g(y_i - y_f)} = \sqrt{2(9.8)(10.0 - 5.0)} = 9.90 \text{ m/s}
 \end{aligned}$$

Find the diver's speed at the water's surface, $y_f = 0$:

$$\begin{aligned}
 0 &= mgy_i = \frac{1}{2}mv_f^2 + 0 \\
 v_f &= \sqrt{2gy_i} = \sqrt{2(9.8)(10.0)} = 14.0 \text{ m/s}
 \end{aligned}$$

6.3.7 Power

Power can be defined as:

- i. The rate at which work is done
- ii. The rate at which energy is transformed from one form to another
- iii. The rate at which energy is transferred from one point to another in a medium

Average amount of power can be found by:

$$P = \frac{\text{Work done}}{\text{Time}} = \frac{\text{Energy transformed}}{\text{time}} \quad (6.18)$$

Example 6.5

A 70.0 kg jogger runs up a long flight of stairs in 4.0 s. the vertical height of the stairs is 4.5 m. (a) estimate the jogger's power output in watts (b) how much energy did this require?

Solution

$$P = \frac{W}{t} = \frac{mgy}{t} = \frac{(70.0 \text{ kg})(9.8 \text{ m/s}^2)(4.5 \text{ m})}{4.0 \text{ s}} = 770 \text{ W}$$

The energy required is:

$$E = Pt = 770 \text{ J/s}(4.0 \text{ s}) = 3100 \text{ J}$$

Unit 7

Linear Momentum

7.3.1 Linear Momentum

Linear momentum is mass multiply by velocity given by:

$$\mathbf{P} = m\mathbf{v} \quad (7.01)$$

Where $\mathbf{P}$ is linear momentum, m is mass and v the object's velocity.

The unit of momentum is kgms^{-1} and it is a vector quantity.

Example 7.1

An object of mass 500g accelerates uniformly from rest at 4ms^{-2} . Calculate its momentum at time $t = 6\text{s}$

Solution:

You need first of all find the final velocity of an object at $t = 6\text{s}$:

Applying equation; $v = u + at$ and noting that initial velocity is $u = 0$ (it accelerates from rest), then you get:

$$\begin{aligned} v &= 0 + 4\text{ms}^{-2}(6\text{s}) \\ v &= 24\text{ms}^{-2} \end{aligned}$$

From the definition of momentum:

$$\begin{aligned} P &= mv \\ P &= 0.5\text{kg}(24\text{ms}^{-1}) \\ P &= 12\text{ms}^{-1} \end{aligned}$$

7.3.2 Linear Momentum and Newton's second law of motion

Newton stated the second law of motion as the resultant force acting on an object is directly proportional to the time rate of change of linear momentum.

We now know that Linear momentum is mass multiply by velocity:

$$\mathbf{P} = m\mathbf{v} \quad (7.01)$$

Where $\mathbf{P}$ is linear momentum, m is mass and v the object's velocity.

Suppose an object of mass m , moving with initial velocity u and changes its velocity to v in time t , then the change of momentum is given by:

$$\Delta\mathbf{P} = m(\mathbf{v} - \mathbf{u}) \quad (7.02)$$

and the time rate of change of momentum becomes:

$$\frac{\Delta\mathbf{P}}{t} = \frac{m(\mathbf{v} - \mathbf{u})}{t}$$

From the definition of Newton's second law in terms of momentum:

$$\mathbf{F}_R \propto m\left(\frac{v-u}{t}\right) \quad (7.03)$$

Relationship 7.03 in equation form becomes:

$$\mathbf{F}_R = \frac{km(v-u)}{t} \quad (7.04)$$

where k is the proportionality constant.

Recall that the first equation of motion is $v = u + at$. Substituting this equation into equation 6.04 for v and taking $k = 1$, you get:

$$\mathbf{F}_R = \frac{km(\mathbf{u} + \mathbf{a}t - \mathbf{u})}{t}$$

which reduces to:

$$\mathbf{F}_R = m\mathbf{a} \quad (7.05)$$

which is the same as equation 4.4 in unit 4.

You notice that the two statements of Newton's second law of motion are the same:

- i. Acceleration is equal to force divided by mass of an object.
- ii. The resultant force acting on an object is directly proportional to the time rate of change of linear momentum in the direction of the force.

7.3.3 The law of Conservation of Linear Momentum

It states that in a closed system, the total momentum of the system remains constant.

To show that linear momentum is conserved, let's consider Figure **Figure 7.1** in which two objects are involved in collision:

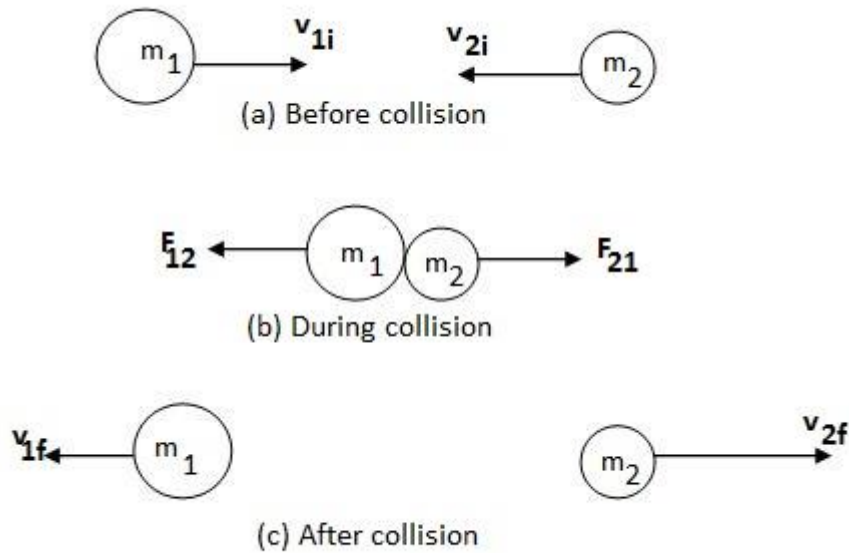


Figure 7.1 Conservation of Momentum

The force experienced by m_1 due to m_2 is F_{12} and the force experienced by m_2 due to m_1 is F_{21} .

Newton's third law of motion tells us that:

$$F_{12} = -F_{21} \quad (7.06)$$

Suppose that m_1 changes its velocity from initial velocity v_{1i} to final velocity v_{1f} during the time interval of Δt and object m_2 changes its velocity from initial value v_{2i} to final value v_{2f} during the same time interval of Δt , then according to Newton's second law:

$$F_{12} = \frac{m_1(v_{1i} - v_{1f})}{\Delta t} \quad (7.07)$$

$$F_{21} = \frac{m_2(v_{2i} - v_{2f})}{\Delta t} \quad (7.08)$$

Substituting 7.06 and 6.08 into 7.06 we get:

$$\begin{aligned} \frac{m_1(v_{1i} - v_{1f})}{\Delta t} &= -\frac{m_2(v_{2i} - v_{2f})}{\Delta t} \\ m_1 v_{1i} - m_1 v_{1f} &= -m_2 v_{2i} + m_2 v_{2f} \end{aligned}$$

and putting initial values and final values to the left and to the right hand sides respectively, we get:

$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f} \quad (7.09)$$

Equation 7.09 is the **Law of Conservation of linear momentum**.

The **law of conservation of linear momentum** states that for an isolated system of objects, the total linear momentum before collision is equal to the total linear momentum after collision.

7.3.4 Impulse

Impulse is the product of the net force and the time interval for which the force acts.

Impulse is defined as:

$$\text{Impulse} = \mathbf{F}_{\text{net}} \cdot \Delta t \quad (7.10)$$

From Newton's second law of motion:

$$\begin{aligned} \mathbf{F}_{\text{net}} &= \frac{\Delta \mathbf{P}}{\Delta t} \\ \mathbf{F}_{\text{net}} \cdot \Delta t &= \Delta \mathbf{P} \end{aligned}$$

Therefore,

$$\text{Impulse} = \Delta \mathbf{P} \quad (7.11)$$

Equation 7.11 is the **impulse-momentum theorem**.

Impulse-momentum theorem states that impulse is equal to the change of an object's linear momentum.

Example 7.2

A 150 N resultant force acts on 300 kg trailer. Calculate how long it takes this force to change the trailer's velocity from 2ms^{-1} to 6ms^{-1} in the same direction.

Solution:

$$m = 300\text{kg}, F_n = 150\text{ N}, u = 2\text{ms}^{-1} \text{ and } v = 6\text{ms}^{-1}$$

$$F_n \cdot \Delta t = \Delta P$$

$$F_n \cdot \Delta t = m(v - u)$$

$$\Delta t = \frac{m(v-u)}{F_n}$$

$$\Delta t = \frac{300\text{kg}(6\text{ms}^{-1} - 2\text{ms}^{-1})}{150\text{ N}} = 8\text{s}$$

7.3.5 Collision in two dimensions

So far you have looked at collisions in one dimension. You should be aware that there are also collisions in two dimensions which objects move off at some angle after collision. This is also called **glancing collision**.

Consider the glancing collision between two billiard balls shown in figure 8.14. Ball 1 is moving to the right at the velocity $\mathbf{v}_{1i}$ and ball 2 is at rest ($\mathbf{v}_{2i} = 0$). After the collision, ball 1 is found to be moving at an angle 45.0° above the horizontal and ball 2 is moving at an angle $\alpha = 45.0^\circ$ below the horizontal. Let us find the velocities of both balls after the collision.

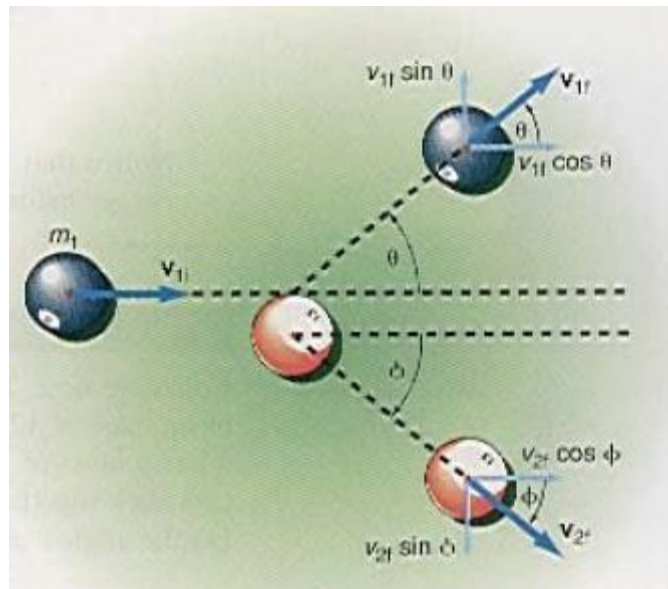


Figure 3.3.4 Glancing collision

As in all collisions, the law of conservation of momentum holds, that is,

$$\mathbf{P}_f = \mathbf{P}_i$$

$$\sum P_x = m_1 v_{1f} \cos \theta + m_2 v_{2f} \cos \vartheta = m_1 v_{1i} \quad (7.12)$$

$$\sum P_y = m_1 v_{1f} \sin \theta - m_2 v_{2f} \sin \vartheta = 0 \quad (7.13)$$

Solving equation 6.20 for v_{2f} , with $\theta = \vartheta = 45.0^\circ$, we get:

$$m_1 v_{1f} \sin 45.0^\circ = m_2 v_{2f} \sin 45.0^\circ$$

$$v_{2f} = \frac{m_1 v_{1f}}{m_2} \quad (7.14)$$

Substituting equation (6.14) into equation 6.12 for v_{2f} and making v_{1f} the subject of the formula, you get:

$$v_{1f} = \frac{v_{1i}}{2 \cos 45.0^\circ} \quad (7.15)$$

Activity 7.1

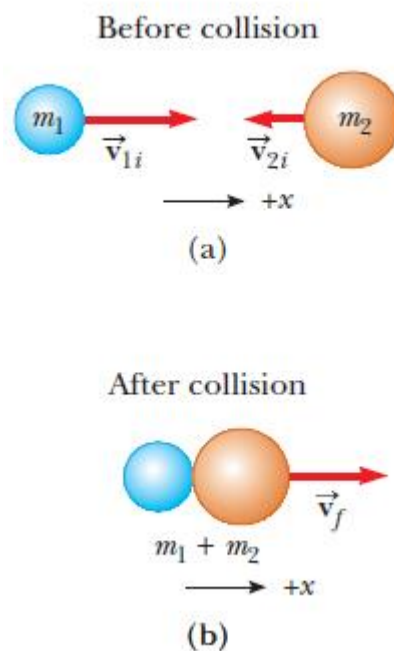
Billiard ball 1 is moving at a speed of $v_{1i} = 10.0 \text{ cm/s}$, when it has a glancing collision with an identical billiard ball that is at rest. After the collision, $\theta = \vartheta = 45.0^\circ$. The mass of the billiard ball is 0.170 kg .

- Find the speed of ball 1 and 2 after the collision.
- Is energy conserved in this collision?

7.3.6 Three types of Collisions

- (i). **Elastic collision** is a collision in which both momentum and kinetic energy are conserved
- (ii). **Inelastic collision** is the one in which momentum is conserved but kinetic energy is not conserved
- (iii). **Perfectly inelastic collision** is the one in which momentum is conserved but kinetic energy is not conserved and the colliding objects stick together and move with the same final velocity after collision.

7.3.6.1 Perfectly Inelastic Collision



Consider two objects of masses m_1 and m_2 moving with initial velocities v_{1i} and v_{2i} respectively before collision. They collide and move with common velocity v_f as shown in the figure above. You can find the common velocity using conservation of momentum alone:

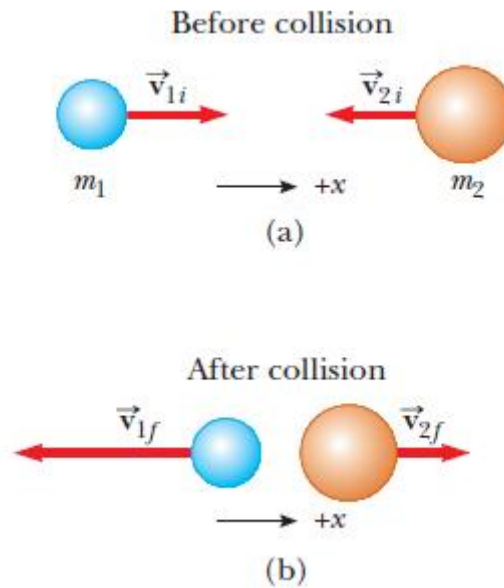
$$P_i = P_f$$

$$m_1 v_{1i} + m_2 v_{2i} = (m_1 + m_2) v_f \quad (7.16)$$

$$v_f = \frac{m_1 v_{1i} + m_2 v_{2i}}{m_1 + m_2} \quad (7.17)$$

7.3.6.2 Elastic Collisions

Let's now consider the collision of two objects in which both momentum and kinetic energy are conserved as shown in the Figure below:



Conservation of momentum is given by:

$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f} \quad (7.18)$$

Conservation of kinetic energy is given by:

$$\frac{1}{2} m_1 v_{1i}^2 + \frac{1}{2} m_2 v_{2i}^2 = \frac{1}{2} m_1 v_{1f}^2 + \frac{1}{2} m_2 v_{2f}^2 \quad (7.19)$$

Cancelling $\frac{1}{2}$ and putting together those terms containing m_1 and those containing m_2 , equation 6.19 becomes:

$$m_1(v_{1i}^2 - v_{1f}^2) = m_2(v_{2f}^2 - v_{2i}^2) \quad (7.20)$$

Factorizing equation 6.20 we get:

$$m_1(v_{1i} - v_{1f})(v_{1i} + v_{1f}) = m_2(v_{2f} - v_{2i})(v_{2f} + v_{2i}) \quad (7.21)$$

You can rewrite equation 6.18 as:

$$m_1(v_{1i} - v_{1f}) = m_2(v_{2f} - v_{2i}) \quad (7.22)$$

Dividing equation 6.21 by Equation 6.22 produces:

$$v_{1i} + v_{1f} = v_{2f} + v_{2i}$$

Gathering initial and final terms together:

$$v_{1i} - v_{2i} = v_{2f} - v_{1f}$$

which can be rewritten as:

$$v_{1i} - v_{2i} = -(v_{1f} - v_{2f}) \quad (7.23)$$

Equation 7.23 together with Equation 7.18 can be used to solve problems dealing with perfectly elastic head-on collisions.

Unit 8

Uniform Circular Motion and Newton's law of Universal Gravitation

8.3 Uniform Circular motion

If a body moves in a circle or a circular arc with constant linear speed, it is said to undergo uniform circular motion. The magnitude of its velocity remains the same but changes direction continuously.

8.3.1 Angular displacement

Consider an object moving in a circular path from position A to position B in time Δt , it starts at an angle θ_i to an angle θ_f as in the diagram below:

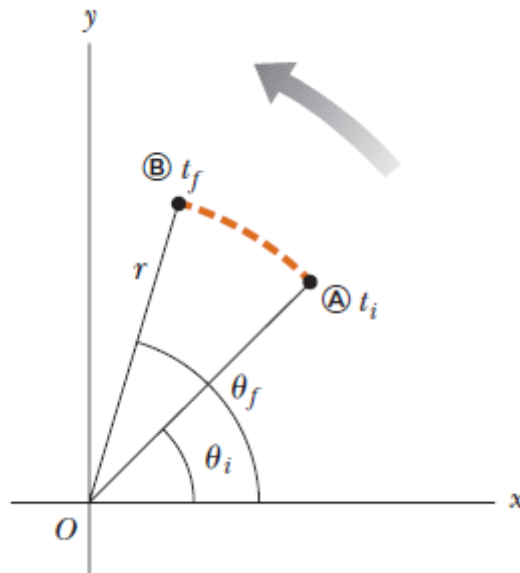


Figure 7.1

Then an object's angular displacement $\Delta\theta$ is the difference in its final and initial angles given by:

$$\Delta\theta = \theta_f - \theta_i \quad (8.01)$$

The SI unit of angular displacement is radian (rad)

From figure 8.1, an object covers a displacement along the circular arc equal to Δs with radius r , then angular displacement can also be expressed by:

$$\Delta\theta = \frac{\Delta s}{r} \quad (8.02)$$

The average angular speed ω_{av} of a rotating rigid object during the time interval Δt is the angular displacement divide by time interval:

$$\omega_{av} = \frac{\Delta\theta}{\Delta t} = \frac{\theta_f - \theta_i}{t_f - t_i} \quad (7.03)$$

The SI unit of angular speed is radian per second (rad/s)

Instantaneous angular speed ω of a rotating rigid object is the limit of average speed

$\frac{\Delta\theta}{\Delta t}$ as time interval Δt approaches zero:

$$\omega = \lim_{\Delta t \rightarrow 0} \frac{\Delta\theta}{\Delta t} \quad (8.04)$$

Its SI unit is radian per second squared (rad/s)

An average angular acceleration α_{av} during the time interval Δt is the change in its angular speed $\Delta\omega$ divide by Δt :

$$\alpha_{av} = \frac{\Delta\omega}{\Delta t} = \frac{\omega_f - \omega_i}{t_f - t_i} \quad (8.05)$$

Its SI unit is radian per second squared (rad/s²)

The instantaneous angular acceleration α is the limit of the average angular acceleration as time interval approaches to zero:

$$\alpha = \lim_{\Delta t \rightarrow 0} \frac{\Delta\omega}{\Delta t} \quad (8.06)$$

Its SI unit is radian per second squared (rad/s²)

Activity 8.1

1. Define the following terms:
 - i. Angular velocity
 - ii. Angular acceleration
 - iii. Instantaneous angular velocity
 - iv. Instantaneous angular acceleration
2. Convert 270° into radian
3. A spot on centrifuge motor moves in a circular path of radius 0.2 m. when the spot has travelled a linear distance of 3 m, through what angle has the motor rotated? Write your answer in radians.

8.3.2 Rotational Kinematics

Rotational kinematics describes rotational motion in the same way linear kinematics was described in unit 2. Remember that we are considering acceleration to be constant both rotational and translation kinematics in this module. The equations of rotational motion can be derived in the same way as you derived the equations of motion in translational motion.

For example, consider a rotating object changing its angular speed from ω_i to ω_f during time interval $\Delta t = t$ i.e during time interval $t_i = 0$ to $t_f = t$., then:

$$\alpha = \frac{\omega_f - \omega_i}{t_f - t_i}$$

$$\alpha = \frac{\omega_f - \omega_i}{t}$$

$$\omega_f = \omega_i + \alpha t \quad (8.07)$$

You see that equation 8.07 is similar to equation 2.01 in unit 2 which is

$$v = u + at$$

You can therefore derive the other equations of rotational motion in the same way as you did in chapter 2. We summarize these equations in the Table 7.01

Table 8.01: Linear equations versus angular equations

Linear equations with $a = \text{constant}$

$$v = u + at$$

$$\Delta s = ut + \frac{1}{2}at^2$$

$$\Delta s = \left(\frac{u + v}{2}\right)t$$

$$v^2 = u^2 + 2a\Delta s$$

Angular acceleration with $a = \text{constant}$

$$\omega_f = \omega_i + \alpha t$$

$$\Delta \theta = \omega_i t + \frac{1}{2}\alpha t^2$$

$$\Delta \theta = \left(\frac{\omega_i + \omega_f}{2}\right)t$$

$$\omega_f^2 = \omega_i^2 + 2\alpha\Delta \theta$$

The angular equations will be used in the same way equations of linear motion were used in solving problems in unit 2.

Example 8.1

A wheel rotates with a constant angular acceleration of 3.50 rad/s^2 . If the angular speed of the wheel is 2.00 rad/s at $t_i = 0$,

- Through what angle does the wheel rotate between $t_i = 0$ and $t_f = 2.00 \text{ s}$?
- What is the angular speed of the wheel at $t = 2.00 \text{ s}$?

Solution

- $\alpha = 3.50 \text{ rad/s}^2$, $\omega_i = 2.00 \text{ rad/s}$ and $t = 2.00 \text{ s}$

$$\Delta \theta = \omega_i t + \frac{1}{2}\alpha t^2$$

$$\Delta \theta = 2.00 \frac{\text{rad}}{\text{s}(2.00 \text{ s})} + \frac{1}{2} \left(\frac{3.50 \text{ rad}}{\text{s}^2} \right) (2.00)^2$$

$$\Delta\theta = 11.0 \text{ rad}$$

$$(b) \omega_f = \omega_i + \alpha t$$

$$\omega_f = 2.00 \text{ rad/s} + (3.5 \text{ rad/s}^2)(2.00 \text{ s})$$

$$\omega_f = 9.00 \text{ rad/s}$$

Example 8.2

An airplane propeller slows from an initial angular speed of 12.5 rev/s to final angular speed of 5.00 rev/s. during this process the propeller rotates through 21.0 revolutions. Find the angular acceleration of the propeller in radians per second squared, assuming its constant.

Solution

$$\Delta\theta = 21.0 \text{ rev} = (21.0)(2\pi \text{ rad}) = 42.0 \text{ rad}$$

$$\omega_i = 12.5 \frac{\text{rev}}{\text{s}} = (12.5) \left(2\pi \frac{\text{rad}}{\text{s}} \right) = 25.0 \text{ rad/s}$$

$$\omega_f = 5.00 \frac{\text{rev}}{\text{s}} = (5.00) \left(2\pi \frac{\text{rad}}{\text{s}} \right) = 10.0\pi \text{ rad/s}$$

$$(10.0\pi \text{ rad/s})^2 = (25.0\pi \text{ rad/s})^2 + 2\alpha(42.0 \text{ rad})$$

$$\alpha = -6.25\pi \text{ rad/s}^2$$

8.3.3 Relationship between angular and linear quantities

Angular variables are closely related to linear variables. Consider an object rotating about the z-axis through the point O. assume the object rotates through an angle $\Delta\theta$, and hence point P moves through the arc length Δs , in time interval Δt . From the definition of radian measure:

$$\Delta\theta = \frac{\Delta s}{r}$$

Dividing both sides of this equation by Δt , we get:

$$\frac{\Delta\theta}{\Delta t} = \frac{\Delta s}{r\Delta t} \quad (8.08)$$

When Δt is very small, the angle $\Delta\theta$ is also small and $\frac{\Delta\theta}{\Delta t}$ is close to instantaneous angular velocity ω where as $\frac{\Delta s}{\Delta t}$ approaches to instantaneous linear velocity v , then equation 8.08 is equivalent to:

$$\omega = \frac{v}{r} \quad (8.09)$$

The magnitude of $\mathbf{v}$ is the linear speed $v = v_t$ called the tangential speed of a particle moving in circular path written as:

$$v_t = r\omega \quad (8.10)$$

If the object changes its angular speed by $\Delta\omega$ in the time interval Δt , this means that tangential speed changes by Δv_t . From equation 7.10, you have:

$$\Delta v_t = r\Delta\omega$$

Dividing both sides of this equation by Δt , you have:

$$\frac{\Delta v_t}{\Delta t} = r \frac{\Delta\omega}{\Delta t} \quad (8.11)$$

As the time interval approaches zero, $\frac{\Delta\omega}{\Delta t}$ approaches instantaneous angular acceleration α and $\frac{\Delta v_t}{\Delta t}$ approaches to instantaneous tangential acceleration a_t . Then equation 8.11 becomes:

$$a_t = r\alpha \quad (8.12)$$

Example 8.3

A compact disc rotates from rest up to an angular speed of 31.4 rad/s in time 0.892 s.

- What is the angular acceleration of the disc, assuming the angular acceleration is uniform?
- Through what angle does the disc turn while coming up to speed?
- If the radius of the disc is 4.45 cm, find the tangential of microbe riding on the rim of the disc when $t=0.892$ s.
- What is the magnitude of the tangential acceleration of the microbe at the given time?

Solution

$$(a) \omega_i = 0, \omega_f = 31.4 \frac{rad}{s} \text{ and } t = 0.892 \text{ s}$$

$$\omega_f = \omega_i + \alpha t$$

$$31.4 \frac{rad}{s} = 0 rad/s + \alpha(0.892 \text{ s})$$

$$\alpha = 35.2 \text{ rad/s}^2$$

$$(b) \Delta\theta = \omega_i t + \frac{1}{2}\alpha t^2$$

$$\Delta\theta = 0(0.892) + \frac{1}{2}(35.2 \frac{rad}{s^2})(0.892 \text{ s})^2$$

$$\Delta\theta = 14.0 \text{ rad}$$

$$(c) v_t = r\omega$$

$$v_t = (0.0445 \text{ m})(31.4 \text{ rad/s})$$

$$v_t = 1.40 \text{ m/s}$$

$$(d) a_t = r\alpha$$

$$a_t = 0.0445 \text{ m})(35.2 \text{ rad/s})$$

$$a_t = 1.57 \text{ m/s}^2$$

8.3.4 Centripetal acceleration

In uniform circular motion, acceleration of an object is always perpendicular to the path and always points towards the centre of the circular path. It is called centripetal acceleration. ‘Centripetal’ means centre seeking. The magnitude of centripetal acceleration is given by:

$$a_c = \frac{v^2}{r} \quad (8.13)$$

Where a_c is centripetal acceleration, v is the linear speed of the object, and r is the radius of the circular path. Centripetal acceleration is also called radial acceleration because its direction coincides with the radius of the circular path at every point.

Centripetal acceleration is also given by:

$$a_c = r\omega^2 \quad (8.14)$$

8.3.5 Centripetal force

The force that causes centripetal acceleration of an object undergoing circular motion is called centripetal force. It is this force that enables an object to move in a circular path.

For an object of mass m , moving in a circular path of radius r , with constant linear speed v , the magnitude of its centripetal force F_c , is given by Newton’s second law of motion:

$$F_c = ma_c = m \frac{v^2}{r} \quad (8.15)$$

When an object moves in a circle but is speeding up or slowing down, a tangential component of acceleration $a_t = r\omega$, is also present. Because the tangential and centripetal components of acceleration are perpendicular to each other, we can find the magnitude of the **total acceleration** with the Pythagorean Theorem:

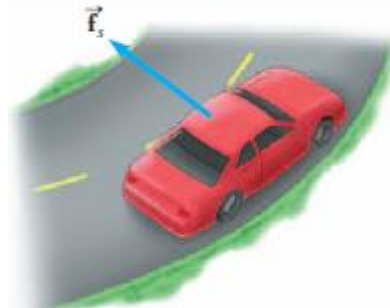
$$a_R = \sqrt{a_t^2 + a_c^2} \quad (8.16)$$

The direction of the resultant acceleration is found by :

$$\Phi = \tan^{-1}(a_c/a_t) \quad (8.17)$$

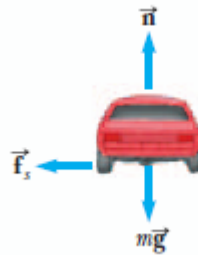
Example 8.4

A car travels at a constant speed of 30.0 mi/h (13.4 m/s) on a level circular turn of radius 50.0 m, as shown in the bird's-eye view in Figure 7.13a. What minimum coefficient of static friction, μ_s , between the tires and roadway will allow the car to make the circular turn without sliding?



Solution:

The net force acting on the car in the radial direction is the force of static friction toward the centre of the circular path which causes the car to have centripetal acceleration.



$$n - mg = 0$$

$$n = mg$$

$$m \frac{v^2}{r} = f_{s \max} = \mu_s n = \mu_s mg$$

$$\mu_s = \frac{v^2}{rg} = \frac{(13.4 \text{ m/s})^2}{(50.0 \text{ m})(9.80 \text{ m/s}^2)} = 0.366$$

The value of μ_s for rubber on dry concrete is very close to 1, so the car can negotiate the curve with ease. If the road were wet or icy, however, the value for μ_s could be 0.2 or lower. Under such conditions, the radial force provided by static friction wouldn't be great enough to keep the car on the circular path, and it would slide off on a tangent, leaving the roadway.

Banking of curved roads and tracks

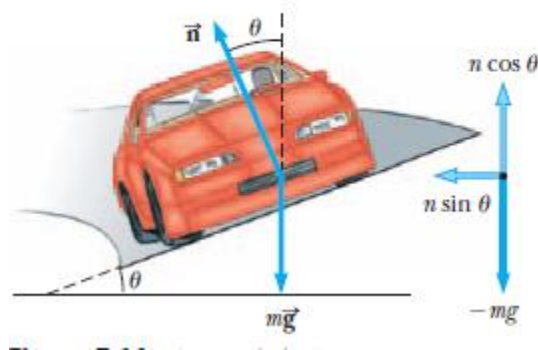
When a car goes round a level curve, the force of friction between the tyres and the road provides the necessary centripetal force. If the frictional force, which acts as centripetal force and keeps the body moving along the circular road is not enough to

provide the necessary centripetal force, the car will skid. In order to avoid skidding, while going round a curved path the outer edge of the road is raised above the level of the inner edge. This is known as banking of curved roads or tracks.

Example 8.5

The Daytona International Speedway in Daytona Beach, Florida, is famous for its races, especially the Daytona 500, held every spring. Both of its courses feature four-story, 31.0° banked curves, with maximum radius of 316 m. If a car negotiates the curve too slowly, it tends to slip down the incline of the turn, whereas if it's going too fast, it may begin to slide up the incline.

- Find the necessary centripetal acceleration on this banked curve so the car won't slip down or slide up the incline. (Neglect friction.)
- Calculate the speed of the race car.



Solution

- From the free-fall diagram:

$$n \cos \theta = mg$$

and

$$n = \frac{mg}{\cos \theta}$$

Centripetal force is given by

$$F_c = ma_c$$

This force is equal to $n \sin \theta = \frac{mg}{\cos \theta} \sin \theta = mg \tan \theta$

Equating the two equations, we get:

$$mg \tan \theta = ma_c$$

$$a_c = g \tan \theta = (9.8 \text{ m/s}^2)(\tan 31.0^\circ) = 5.89 \text{ m/s}^2$$

- We can get the speed of the race car by:

$$a_c = \frac{v^2}{r} \rightarrow v = \sqrt{ra_c} = \sqrt{316 \times 5.89} = 43.14 \text{ ms}^{-1}$$

Activity 8.1

1. A 1.50-kg bucket of water is tied by a rope and whirled in a circle with a radius of 1.00 m. At the top of the circular loop, the speed of the bucket is 4.00 m/s.

Determine the acceleration, the net force and the individual force values when the bucket is at the top of the circular loop

9.3.6 Newton's law of universal gravitation

Newton's law of universal gravitation states that: Every particle in the universe attracts every other particle with a force that is proportional to the product of their masses and inversely proportional to the square of the distance between them. This force acts along the line joining the two particles.

If two particles having masses m_1 and m_2 are separated by a distance r , a gravitational force F acts along a line joining them, with magnitude given by

$$F = G \frac{m_1 m_2}{r^2} \quad (9.18)$$

Where $G = 6.67 \times 10^{-11} \text{ kg}^{-1} \text{ m}^3 \text{ s}^{-2}$, is proportionality constant called the constant of universal gravitation. The gravitational force is always attractive.

Example:

A 50 kg Physics student and a 75 Physics teacher are sitting on a beach so that their centers are about 50 cm apart. Estimate the magnitude of gravitational force each exerts on the other.

Solution:

$$m_1 = 50 \text{ kg}, m_2 = 75 \text{ kg}, r = 50 \text{ cm} = 0.50 \text{ m}, \text{ and } G = 6.67 \times 10^{-11} \text{ kg}^{-1} \text{ m}^3 \text{ s}^{-2}$$

$$\text{Then: } F = G \frac{m_1 m_2}{r^2} = \frac{6.67 \times 10^{-11} \text{ kg}^{-1} \text{ m}^3 \text{ s}^{-2} \times 50 \text{ kg} \times 75 \text{ kg}}{(0.50 \text{ m})^2} = 1.0 \times 10^{-6} \text{ N} \text{ This is unnoticeably small.}$$

In everyday life experience, the gravitational force between two objects is not felt because the masses involved are too small to produce an appreciable value of attractive force.

NOTE

The law of universal gravitation describes a particular force, gravity, and how its magnitude varies with the distance and masses involved.

Newton's second law of motion relates the net force acting on an object to the mass and acceleration of that object.

9.3.7 Gravitational force and weight

Weight is the earth's gravitational force acting on a particular object. In the Equation of the Law of universal gravitation, if one of the object is the earth having mass, M_E , then gravitational force is given by:

$$F_g = mg = G \frac{M_E m}{r^2} \quad (9.19)$$

where g is the acceleration due to gravity.

From Equation 8.19 you can make g the subject of the formula to get:

$$g = G \frac{M_E}{r^2} \quad (9.20)$$

With $r = R_E + h$ where R_E is the radius of the earth and h is the height above sea level.

In equation 9.20, G and M_E are constants, therefore, g varies inversely as r^2 only. If r approaches infinity, g approaches zero. As a result the weight of the object also approaches to zero. This explains the reason why in the outer space object becomes weightless.

Activity 8.2

1. Explain the differences between mass and weight
2. Three 4-kg masses are each at a corner of an equilateral triangle located in space far from other masses.
 - a. If the sides of the triangle are 1.00 m long, find the magnitude of the net force exerted on each of the three masses.
 - b. How does the net force change if the sides of the triangle are halved in length?